

SGA 7-I

Exposé VII continuation through Exposé IX §6.6

Biextensions and the pairings associated with them

English translation

Exposé VII. Biextensions of sheaves of groups, continued

2.3. Morphisms of biextensions

Let E be a biextension of (P, Q) by G , and likewise let E' be a biextension of (P', Q') by G' , where P', Q' , and G' are again commutative groups of the topos $\mathcal{T}$. A *morphism of biextensions* from E to E' is a system

$$(u, v, w, f)$$

where

$$u : P \longrightarrow P', \quad v : Q \longrightarrow Q', \quad w : G \longrightarrow G'$$

are homomorphisms of groups, and where

$$f : E \longrightarrow E'$$

is a morphism of the underlying sheaves of sets, such that f is at the same time a homomorphism of extensions over P , respectively over P' , associated with the corresponding homomorphisms of relative groups, and a homomorphism of extensions over Q , respectively over Q' , associated with the corresponding homomorphisms of relative groups. In other words the two group-extension structures carried by a biextension must be respected simultaneously.

The maps u, v, w are already determined by f . Passing to the quotient recovers $u \times v$, hence u and v by restriction to $P \times 1_Q$ and $1_P \times Q$. Restricting f to the copy of G through the neutral section recovers w . Thus no ambiguity results from identifying the morphism (u, v, w, f) with the single map f .

With the usual precaution of recording source and target, biextensions in $\mathcal{T}$ are therefore the objects of a category. We denote it by **BIEXT**($\mathcal{T}$). The assignment

$$(E; P, Q, G) \longmapsto (P, Q, G)$$

defines a canonical functor

$$\mathbf{BIEXT}(\mathcal{T}) \longrightarrow \mathbf{GRAB}(\mathcal{T}) \times \mathbf{GRAB}(\mathcal{T}) \times \mathbf{GRAB}(\mathcal{T}),$$

where $\mathbf{GRAB}(\mathcal{T})$ denotes the category of abelian groups in $\mathcal{T}$. It is often useful to decompose this functor into the two functors

$$E \longmapsto (P, Q), \quad E \longmapsto G.$$

2.4. The category $\mathbf{BIEXT}(P, Q; G)$ with P, Q, G variable

The first of the two preceding functors is fibred, and the second is cofibrant. Concretely, if E is a biextension of (P, Q) by G , and if

$$u : P_1 \rightarrow P, \quad v : Q_1 \rightarrow Q$$

are homomorphisms of abelian groups, then there is a biextension E' of (P_1, Q_1) by G and a morphism of biextensions

$$E' \longrightarrow E$$

compatible with u, v, id_G , having the usual universal property. This biextension is the inverse image of E by (u, v) .

Dually, if $w : G \rightarrow G'$ is a homomorphism of abelian groups, there is a biextension E' of (P, Q) by G' and a morphism of biextensions

$$E \longrightarrow E'$$

compatible with $\text{id}_P, \text{id}_Q, w$, again characterized by its universal property. It is the push-out of E by w . These constructions are obtained by paraphrasing the corresponding constructions for ordinary extensions of groups.

For fixed P, Q, G we write

$$\mathbf{BIEXT}(P, Q; G)$$

for the fibre category over the triple (P, Q, G) . The preceding description gives the usual functorialities:

$$(u, v)^* : \mathbf{BIEXT}(P, Q; G) \rightarrow \mathbf{BIEXT}(P_1, Q_1; G),$$

and

$$w_* : \mathbf{BIEXT}(P, Q; G) \rightarrow \mathbf{BIEXT}(P, Q; G').$$

These are compatible in the evident sense.

2.5. The structure of $\mathbf{BIEXT}(P, Q; G)$

For fixed P, Q, G , the category $\mathbf{BIEXT}(P, Q; G)$ is a strictly commutative Picard category. The sum of two biextensions is defined by taking the contracted product over $P \times Q$:

$$E + E' = (E \times_{P \times Q} E')/G,$$

where G is embedded diagonally with the usual opposite action on one factor. The two extension laws on E and E' induce the two required extension laws on $E + E'$, and the compatibility axiom for a biextension is inherited from the corresponding axioms for E and E' .

The neutral object is the trivial biextension

$$P \times Q \times G$$

with its evident partial addition laws. The inverse of a biextension is obtained by replacing the G -torsor E by its opposite torsor. Associativity and commutativity are inherited from the corresponding facts for contracted products of torsors. Thus the isomorphism classes form an abelian group, denoted

$$\mathrm{Biext}^1(P, Q; G).$$

The automorphism group of the neutral biextension is also useful. An automorphism of the trivial biextension over $P \times Q$ is given by a morphism

$$P \times Q \longrightarrow G.$$

The condition that it respect both extension structures says precisely that this morphism is additive separately in P and in Q . Thus

$$\mathrm{Aut}_{\mathbf{BIEXT}(P, Q; G)}(0) \simeq \mathrm{Bilin}(P, Q; G),$$

the group of biadditive morphisms $P \times Q \rightarrow G$. We denote this group by

$$\mathrm{Biext}^0(P, Q; G).$$

The two groups Biext^0 and Biext^1 behave functorially in all three arguments. Pulling back in P and Q , and pushing out in G , gives the corresponding homomorphisms of Picard categories and hence of their homotopy groups.

2.6. Opposite and transposed biextensions

Let E be a biextension of (P, Q) by G . One may form the opposite biextension, obtained by replacing the G -torsor by its inverse. It represents the negative of the class of E in $\text{Biext}^1(P, Q; G)$.

One may also transpose the two variables. The transposed biextension

tE

is a biextension of (Q, P) by G , obtained by pulling E back along the symmetry $Q \times P \simeq P \times Q$ and interchanging the two extension structures. This operation is involutive and is compatible with inverse images and direct images.

2.7. Symmetric biextension associated with a given biextension

The sum of E with its transpose, when the two variables are identified, gives the natural symmetric construction. More generally, whenever a symmetry condition is imposed on P and Q , a biextension may be equipped with an isomorphism to its transpose satisfying the usual cocycle condition. This gives the category of symmetric biextensions. In the applications to pairings and to duality, this is the construction which records the symmetry or alternating behaviour of the pairing arising from a biextension.

2.8. Change of topos. Localization

Let $u : \mathcal{T}' \rightarrow \mathcal{T}$ be a morphism of topoi. If E is a biextension of (P, Q) by G in $\mathcal{T}$, then u^*E is a biextension of (u^*P, u^*Q) by u^*G in $\mathcal{T}'$. This construction is functorial and respects sums, inverses, and transposition.

If X is an object of $\mathcal{T}$, the localization $\mathcal{T}/X$ gives a special case. The restriction of E to X is a biextension of the restrictions of P and Q by the restriction of G . Conversely, assertions about biextensions which are local for the topology of $\mathcal{T}$ may be checked after such localizations. This is used repeatedly below when questions of existence or uniqueness are reduced to the case of a torsor admitting a section.

The formation of $\text{Biext}^i(P, Q; G)$ is therefore compatible with change of topos in the expected way:

$$u^* \text{Biext}^i(P, Q; G) \longrightarrow \text{Biext}^i(u^*P, u^*Q; u^*G).$$

For $i = 0$ this is the usual pullback of biadditive morphisms; for $i = 1$ it is obtained by pulling back representatives.

2.9. Relation with bitrivialized torsors

A biextension may be described equivalently as a torsor over $P \times Q$ under G , together with two compatible partial group laws. If one chooses trivializations along the two axes $P \times 1_Q$ and $1_P \times Q$, agreeing at the neutral section, the remaining datum is often encoded by a torsor equipped with two rigidifications. This is the bitrivialized-torsor description.

This point of view is especially useful for explicit computations. Given a G -torsor E over $P \times Q$, a rigidification along $P \times 1_Q$ and one along $1_P \times Q$ allow one to speak of the defect of additivity in each variable. The compatibility condition for a biextension says exactly that these defects are themselves compatible; in a Čech description it is the usual two-variable cocycle condition.

The equivalence between the two descriptions is natural in P, Q, G , compatible with change of topos, and preserves the Picard structure.

2.10. Variants of the notion of biextension

The definition admits evident variants. One may replace abelian groups by group objects subject to commutativity restrictions sufficient for the two partial extension structures to commute. One may also introduce rigidified, symmetrized, or alternating biextensions, according to whether additional trivializations or symmetry constraints are imposed.

There are also relative variants, in which P, Q, G are objects over a base object of the topos. They are obtained by applying the preceding definitions in the localized topos. All these variants are stable under pullback and push-out and will be used without further comment when the context requires them.

3. Complements of homological algebra

3.1. The additive cofibration $\mathcal{F}_{L_\bullet}$

Let $\mathcal{C}$ be an additive category in which the extensions under consideration are defined, and let

$$L_\bullet : \cdots \longrightarrow L_2 \longrightarrow L_1 \longrightarrow L_0$$

be a chain complex of objects of $\mathcal{C}$. To $L_\bullet$ one associates a cofibred additive category

$$\mathcal{F}_{L_\bullet}$$

over $\mathcal{C}$. The fibre over an object A is the category of pairs consisting of an extension

$$0 \longrightarrow A \longrightarrow X \longrightarrow L_0 \longrightarrow 0$$

together with a lifting of the differential $L_1 \rightarrow L_0$ to X , subject to the condition imposed by the next differential $L_2 \rightarrow L_1$.

More explicitly, an object of the fibre over A is a pair (X, α) , where X is an extension of L_0 by A and

$$\alpha : L_1 \longrightarrow X$$

satisfies

$$j\alpha = d_1, \quad \alpha d_2 = 0$$

with $j : X \rightarrow L_0$ the quotient morphism. Morphisms are morphisms of extensions compatible with α . The resulting category is additive and cofibred in A , by push-out of extensions.

There is a canonical interpretation of its connected components and automorphism groups in terms of Ext groups:

$$\pi_0 \mathcal{F}_{L_\bullet}(A) \simeq \text{Ext}^1(L_\bullet, A), \quad \pi_1 \mathcal{F}_{L_\bullet}(A) \simeq \text{Ext}^0(L_\bullet, A),$$

where the Ext groups are computed from the complex $L_\bullet$.

3.2. Determination of $\mathcal{F}_{L_\bullet}^1(A)$

The group $\mathcal{F}_{L_\bullet}^1(A)$ is the group of isomorphism classes of objects of the fibre over A . The preceding description shows that it is exactly the group of extensions of the complex $L_\bullet$ by A , modulo the usual equivalence relation. In formulas,

$$\mathcal{F}_{L_\bullet}^1(A) \simeq \text{Ext}^1(L_\bullet, A).$$

Under this correspondence, the Baer sum of extensions agrees with the addition in Ext^1 . Likewise, automorphisms of the zero object correspond to

$$\text{Hom}(H_0(L_\bullet), A)$$

or, more generally, to $\text{Ext}^0(L_\bullet, A)$ in the derived sense.

The construction is compatible with morphisms of complexes. A morphism $L'_\bullet \rightarrow L_\bullet$ gives an inverse-image functor

$$\mathcal{F}_{L_\bullet} \longrightarrow \mathcal{F}_{L'_\bullet},$$

and the induced maps on $\mathcal{F}^i$ are the usual maps on Ext groups.

3.3. Exactness properties of $\mathcal{F}_{L_\bullet}$

Suppose a morphism of complexes

$$L'_\bullet \longrightarrow L_\bullet$$

induces isomorphisms on H_0 and H_1 . Then for every A the induced functor

$$\mathcal{F}_{L_\bullet}(A) \longrightarrow \mathcal{F}_{L'_\bullet}(A)$$

is an equivalence of categories. This is the categorical form of the fact that extensions of degree 0 and 1 depend only on the truncation of the complex through H_1 .

For a short exact sequence of complexes

$$0 \longrightarrow L'_\bullet \longrightarrow L_\bullet \longrightarrow L''_\bullet \longrightarrow 0,$$

one obtains the exact sequence

$$0 \rightarrow \text{Ext}^0(L''_\bullet, A) \rightarrow \text{Ext}^0(L_\bullet, A) \rightarrow \text{Ext}^0(L'_\bullet, A) \rightarrow \text{Ext}^1(L''_\bullet, A) \rightarrow \text{Ext}^1(L_\bullet, A) \rightarrow \cdots.$$

The connecting homomorphism admits a direct categorical description: an element of $\text{Ext}^0(L'_\bullet, A)$, viewed as an automorphism of the neutral object in $\mathcal{F}_{L'_\bullet}(A)$, measures the obstruction to descending the corresponding trivialized object along the exact sequence of complexes. This obstruction is the image in $\text{Ext}^1(L''_\bullet, A)$.

The sign conventions are those of the derived category. Since only the resulting Picard categories and exactness assertions will be used later, the possible ambiguities in signs do not affect the applications that follow.

3.4. Variance of $\mathcal{F}_{L_\bullet}$ with respect to $L_\bullet$

Let

$$0 \rightarrow L'_\bullet \rightarrow L_\bullet \rightarrow L''_\bullet \rightarrow 0$$

be an exact sequence of complexes. The functor

$$\mathcal{F}_{L_\bullet} \longrightarrow \mathcal{F}_{[L'_\bullet \rightarrow L''_\bullet]}$$

which sends an extension of $L_\bullet$ to its restriction to $L'_\bullet$, equipped with the induced trivialization after pullback to $L''_\bullet$, is an equivalence of categories. Here $\mathcal{F}_{[L'_\bullet \rightarrow L''_\bullet]}$ denotes the category of objects of $\mathcal{F}_{L'_\bullet}$ together with a trivialization of their inverse image.

Indeed, an object in the target consists of an extension

$$0 \rightarrow A \rightarrow X' \rightarrow L'_0 \rightarrow 0$$

together with the auxiliary maps encoding the liftings from L'_1 and L''_1 , satisfying the evident relations. These data combine uniquely into an object of $\mathcal{F}_{L_\bullet}$, and the construction is inverse to restriction. This proves the asserted equivalence and gives the left-exactness in the complex-variable which complements the left-exactness in A .

The preceding equivalence identifies the connecting homomorphism in the long exact Ext sequence with the geometric operation of taking the object whose trivialization fails to descend. This is the compatibility needed below.

3.5. A canonical partial flat resolution of an arbitrary $\mathbb{Z}$ -module

We apply the preceding results to the category of commutative groups of the topos $\mathcal{T}$, that is, to the category of $\mathbb{Z}$ -modules of $\mathcal{T}$. Let P be such a group. Define a chain complex $L_\bullet(P)$, augmented toward P , by

$$0 \longrightarrow \mathbb{Z}[P \times P \times P] \xrightarrow{d_2} \mathbb{Z}[P \times P] \xrightarrow{d_1} \mathbb{Z}[P] \xrightarrow{\epsilon} P \longrightarrow 0.$$

For a section p of P , write $[p]$ for the corresponding section of the free module $\mathbb{Z}[P]$, and use analogous notation $[p, q]$ and $[p, q, r]$. Then

$$\epsilon[p] = p,$$

$$d_1[p, q] = [p + q] - [p] - [q],$$

and

$$d_2[p, q, r] = [p + q, r] - [p, q + r] + [p, r] - [q, r].$$

These formulas define a complex:

$$\epsilon d_1 = 0, \quad d_1 d_2 = 0.$$

The first identity is immediate. The second decomposes into the identities expressing commutativity and associativity of the group law on P .

The construction is functorial in P . Each component $L_i(P)$ is a free $\mathbb{Z}$ -module, hence flat.

Proposition 0.1. *One has $H_1(L_\bullet(P)) = 0$, and the augmentation induces an isomorphism*

$$H_0(L_\bullet(P)) \xrightarrow{\sim} P.$$

It is enough to prove that for every commutative group G of $\mathcal{T}$, the functor

$$\mathcal{F}_{L_\bullet(P)}(G) \longrightarrow \mathcal{F}_P(G)$$

is an equivalence. The left hand side identifies with the category of extensions of the $\mathbb{Z}$ -module P by G . Interpreting extensions of the free modules $\mathbb{Z}[P]$, $\mathbb{Z}[P \times P]$, and $\mathbb{Z}[P \times P \times P]$ as torsors over P , $P \times P$, and $P \times P \times P$, respectively, an object of the right hand side is a G_P -torsor on P together with a trivialization of its pullback to $P \times P$, satisfying the compatibility imposed by $d_1 d_2 = 0$. This is exactly the classical description of an extension of the group object P by G . The two constructions are inverse to one another.

3.6. Application to the cofibred category of biextensions of a fixed pair

Let P, Q, G be commutative groups in $\mathcal{T}$. Apply the preceding construction to both P and Q , and set

$$L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q).$$

The complex $L_\bullet(P, Q)$ is a flat resolution, in the relevant low degrees, of $P \otimes^{\mathbf{L}} Q$. The general formalism yields a Picard category

$$\mathcal{F}_{L_\bullet(P, Q)}(G).$$

The category $\mathbf{BIEXT}(P, Q; G)$ is canonically equivalent to this Picard category. Under this equivalence, the class of a biextension corresponds to the class of the associated extension datum for the complex $L_\bullet(P, Q)$. Therefore

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

and

$$\mathrm{Biext}^0(P, Q; G) \simeq \mathrm{Ext}^0(P \otimes^{\mathbf{L}} Q, G).$$

The latter is precisely the group of biadditive maps $P \times Q \rightarrow G$.

This description is functorial in G , by push-out, and in P, Q , by pullback. It is compatible with the Picard structure and with change of topos.

3.7. Various compatibilities

The equivalence just described is compatible with the usual exactness properties in each variable. For an exact sequence

$$0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0$$

and fixed Q, G , the sequence of Picard categories of biextensions is left-exact. On isomorphism classes and automorphism groups it gives the beginning of the long exact Ext sequence associated with

$$0 \rightarrow P' \otimes^{\mathbf{L}} Q \rightarrow P \otimes^{\mathbf{L}} Q \rightarrow P'' \otimes^{\mathbf{L}} Q \rightarrow 0.$$

Equivalently, a biextension of (P, Q) whose inverse image to (P', Q) is trivial has an obstruction class in the group of biextensions of (P'', Q) .

Similarly, exactness in Q is obtained by symmetry, and exactness in G is the exactness of push-outs of extensions. The connecting morphisms are the boundary maps in the corresponding Ext sequences.

Thus the geometrically defined group $\text{Biext}^1(P, Q; G)$ is expressed as the simple homological invariant $\text{Ext}^1(P \otimes^{\mathbf{L}} Q, G)$, and all the natural operations on biextensions are the operations induced on Ext.

3.8. Biextensions of (P, Q) trivialized on extensions of P and Q

Consider two exact sequences of abelian groups

$$0 \rightarrow L_1 \xrightarrow{\partial} L_0 \xrightarrow{u} P \rightarrow 0, \quad 0 \rightarrow M_1 \xrightarrow{\partial} M_0 \xrightarrow{v} Q \rightarrow 0.$$

We describe the category of biextensions E of (P, Q) by an abelian group G whose pullback $(u, v)^*E$ is trivial. A trivialization is a section

$$s : L_0 \times M_0 \longrightarrow (u, v)^*E$$

additive in each variable.

Once s is chosen, the two partial restrictions of the trivial biextension give homomorphisms

$$f : L_1 \otimes M_0 \rightarrow G, \quad g : L_0 \otimes M_1 \rightarrow G.$$

They must agree on $L_1 \otimes M_1$, in the sense that

$$f(x_1 \otimes \partial y_1) = g(\partial x_1 \otimes y_1).$$

Conversely, such a pair (f, g) reconstructs the required descent datum for the trivialized biextension.

Changing the trivialization by a bilinear form

$$h : L_0 \otimes M_0 \rightarrow G$$

changes f and g by the restrictions of h :

$$f'(x_1 \otimes y_0) = f(x_1 \otimes y_0) + h(\partial x_1 \otimes y_0),$$

$$g'(x_0 \otimes y_1) = g(x_0 \otimes y_1) + h(x_0 \otimes \partial y_1).$$

Proposition 0.2. *The subgroup of $\text{Biext}^1(P, Q; G)$ consisting of classes whose pullback to (L_0, M_0) is trivial is canonically isomorphic to the quotient of the group of pairs (f, g) , with*

$$f : L_1 \otimes M_0 \rightarrow G, \quad g : L_0 \otimes M_1 \rightarrow G,$$

satisfying the compatibility condition on $L_1 \otimes M_1$, by the subgroup of pairs obtained by restricting a form $h : L_0 \otimes M_0 \rightarrow G$.

Interpreting the two exact sequences as complexes

$$L_\bullet : 0 \rightarrow L_1 \rightarrow L_0 \rightarrow 0, \quad M_\bullet : 0 \rightarrow M_1 \rightarrow M_0 \rightarrow 0,$$

their tensor product is

$$0 \rightarrow L_1 \otimes M_1 \rightarrow L_1 \otimes M_0 \oplus L_0 \otimes M_1 \rightarrow L_0 \otimes M_0 \rightarrow 0.$$

The preceding quotient is precisely

$$H^1(\text{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)).$$

The canonical morphism

$$L_\bullet \otimes M_\bullet \longrightarrow P \otimes^{\mathbf{L}} Q$$

therefore induces

$$H^1(\text{Hom}^\bullet(L_\bullet \otimes M_\bullet, G)) \longrightarrow \text{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

and the isomorphism with $\text{Biext}^1(P, Q; G)$ sends the class described by (f, g) to this image.

In the important special case

$$L_0 = \mathbb{Z}[P], \quad M_0 = \mathbb{Z}[Q],$$

with L_1 and M_1 the kernels of the augmentations to P and Q , the data (f, g) can be written explicitly as functions

$$P \times P \times Q \rightarrow G, \quad P \times Q \times Q \rightarrow G.$$

The five compatibility conditions are exactly the five axioms which define the two partial group laws of a biextension. Thus the explicit cocycle description and the intrinsic definition agree.

Bibliography to Exposé VII

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Exposé VIII. Complements on biextensions; general properties of biextensions of group schemes

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0. Introduction

We continue the general study of biextensions begun in the preceding exposé. We first add several complements in the case of an arbitrary base topos, developing in particular the formalism of pairings associated with biextensions. This formalism is well known to be important in the theory of abelian schemes. We then begin the study of biextensions of general group schemes. The case of greatest interest for the sequel is that of biextensions by the multiplicative group $\mathbb{G}_m$; nevertheless, as technical intermediates, we shall also study biextensions by other types of group schemes.

The most useful result for what follows is the criterion of Section 7: under suitable hypotheses, if P and Q are smooth over a trait with generic point η , a biextension of (P_η, Q_η) by $\mathbb{G}_m$ extends to a biextension of (P, Q) by $\mathbb{G}_m$, uniquely up to a unique isomorphism. This will be convenient in the following exposé, which is devoted to the Néron model of abelian varieties. It is likely that one could proceed there by a shorter route; nevertheless the developments of Exposés VII and VIII are expected to be useful elsewhere as well.

When working with group schemes over a scheme S , the ambient topos is the fppf topos of S , associated with the site of schemes locally of finite presentation over S , endowed with the faithfully flat topology. The word “local” without further qualification means local for this topology.

Most statements about extensions or biextensions by $\mathbb{G}_m$ remain valid with $\mathbb{G}_m$ replaced by a torus G . Since such a torus is locally isomorphic, for the fppf topology and even for the étale topology, to a power $\mathbb{G}_m^r$, the proofs are obtained by descent.

1. Various special cases over an arbitrary topos

In this section, as in Exposé VII, T is a topos and P, Q, G are abelian groups in T . We use the canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G).$$

It gives immediate vanishing and comparison results in a number of special situations.

If

$$\mathrm{Ext}^1(P \otimes Q, G) = 0$$

and

$$\mathrm{Hom}(\mathrm{Tor}_1(P, Q), G) = 0,$$

then every biextension of (P, Q) by G is trivial. More precisely, the spectral sequence for the derived tensor product gives the exact sequence

$$0 \rightarrow \mathrm{Ext}^1(P \otimes Q, G) \rightarrow \mathrm{Biext}^1(P, Q; G) \rightarrow \mathrm{Hom}(\mathrm{Tor}_1(P, Q), G) \rightarrow \mathrm{Ext}^2(P \otimes Q, G),$$

and the assertion follows under the corresponding vanishing hypotheses.

If P is a flat $\mathbb{Z}$ -module, then

$$P \otimes^{\mathbf{L}} Q \simeq P \otimes Q,$$

hence

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes Q, G).$$

The same conclusion holds, by symmetry, if Q is flat. In particular, if both P and Q are locally free abelian sheaves, biextensions are ordinary extensions of $P \otimes Q$ by G .

Suppose P is killed by an integer $n > 0$. The exact triangle associated with multiplication by n on Q relates $P \otimes^{\mathbf{L}} Q$ to $\mathrm{Tor}_1(P, Q)$ and to $P \otimes Q$. This provides the exact sequences used below to extract torsion pairings from a biextension.

When both P and Q are n -torsion, the derived tensor product is controlled by $\mathrm{Tor}_1(P, Q)$ and $P \otimes Q$, and the biextension group is described by the same Ext sequence. These elementary cases will be applied to finite flat group schemes and to the finite layers of Tate modules.

2. Pairings defined by a biextension

Let P and Q be abelian groups of T , and let $n > 0$ be an integer. Write ${}_nP$ for the kernel of multiplication by n on P , and similarly for Q . There is a canonical homomorphism

$${}_nP \otimes {}_nQ \longrightarrow \mathrm{Tor}_1(P, Q).$$

It is obtained from the exact sequence

$$0 \rightarrow {}_nQ \rightarrow Q \xrightarrow{n} Q$$

by applying $P \otimes^{\mathbf{L}} -$. Equivalently, a pair (p, q) , with $np = 0$ and $nq = 0$, defines the standard Tor class represented by the relation $nq = 0$ pulled against p .

One may describe this homomorphism explicitly. For a section q of ${}_nQ$, lift it in the two-term resolution

$$Q \xrightarrow{n} Q.$$

Tensoring with P , a section $p \in {}_nP$ yields a cycle. Its class in $\mathrm{Tor}_1(P, Q)$ is the image of $p \otimes q$. The construction is functorial in both variables and is bilinear.

Interchanging P and Q gives a second construction

$${}_nQ \otimes {}_nP \longrightarrow \mathrm{Tor}_1(Q, P).$$

Under the canonical symmetry

$$\mathrm{Tor}_1(P, Q) \simeq \mathrm{Tor}_1(Q, P),$$

the two constructions differ by the usual sign. This is the standard Koszul sign arising from the symmetry of the derived tensor product. Thus the induced pairing is alternating in the expected graded sense.

Now let G be a further abelian group and let

$$\xi \in \mathrm{Biext}^1(P, Q; G).$$

Through the isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

and the canonical map

$${}_nP \otimes {}_nQ \rightarrow \mathrm{Tor}_1(P, Q),$$

the biextension ξ defines a pairing

$${}_nP \times {}_nQ \longrightarrow G/nG$$

whenever the target is interpreted through the connecting morphism determined by multiplication by n on G . More invariantly, it defines a morphism

$${}_nP \otimes {}_nQ \longrightarrow \mathrm{Coker}(G \xrightarrow{n} G).$$

If $G = \mathbb{G}_m$, this is the familiar Weil-type pairing attached to a biextension.

When n varies through powers of a prime ℓ , the preceding pairings form a compatible projective system. For any commutative group E , write

$$T_\ell(E) = \varprojlim_{\nu} \ell^\nu E.$$

A biextension ξ then defines an ℓ -adic pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(G)$$

or, after the standard shift appropriate to the multiplicative group, a pairing with values in the Tate module of $\mathbb{G}_m$. These pairings are functorial in all variables and compatible with pullback, push-out, and change of topos.

In concrete terms, take local sections $p \in {}_n P$ and $q \in {}_n Q$. Restrict the biextension E to the subgroup generated by p and q . The relations $np = 0$ and $nq = 0$ provide two ways of comparing the neutral section with itself after n successive additions. Their quotient is an element of G/nG . This element is the value of the pairing at (p, q) . The biextension axiom is precisely what ensures that the value is independent of the choices and is bilinear.

2.3. Relation with Cartier duality

The description just given, through the isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \otimes^{\mathbf{L}} Q, G),$$

is not the original definition of the pairing attached to a biextension. In the classical setting of finite locally free commutative group schemes and $G = \mathbb{G}_m$, the pairing is usually described by Cartier duality.

Let P and Q be finite locally free commutative group schemes over a base S , killed by n , and let E be a biextension of (P, Q) by $\mathbb{G}_m$. Fixing a section p of P , the restriction of E to $\{p\} \times Q$ is an extension of Q by $\mathbb{G}_m$, hence determines a section of the Cartier dual $Q^\vee$. In this way E gives a morphism

$$P \longrightarrow Q^\vee.$$

Equivalently it gives a bilinear pairing

$$P \times Q \longrightarrow \mathbb{G}_m.$$

The homological construction above recovers the same morphism. The verification is local on the

base and reduces to the explicit description of the Tor class associated with p and q , together with the concrete computation of the commutator or monodromy element in the restricted biextension. Thus the two viewpoints agree: the pairing obtained by Cartier duality is the pairing obtained from the Ext-class of the biextension by applying the canonical map from torsion to Tor_1 . This compatibility is the bridge used below between the formal theory of biextensions and the duality theory of finite flat group schemes.

2.4. Pairings attached to torsion and passage to limits

Let n and m be positive integers. The constructions of the previous section are compatible with the transition morphisms

$${}_{mn}P \longrightarrow {}_nP, \quad {}_{mn}Q \longrightarrow {}_nQ$$

and with the corresponding transition maps on roots of unity or on the cokernels of multiplication. Thus an element $\xi \in \mathrm{Biext}^1(P, Q; \mathbb{G}_m)$ gives, for every n , a pairing

$${}_nP \times {}_nQ \longrightarrow \mu_n$$

whenever the relevant finite flatness hypotheses hold. These pairings are compatible in n , and hence for every prime ℓ yield a pairing of Tate modules

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1).$$

Equivalently, after tensoring with $\mathbb{Q}_\ell$, one obtains

$$V_\ell(P) \times V_\ell(Q) \longrightarrow \mathbb{Q}_\ell(1).$$

When P and Q are dual abelian schemes, and ξ is the Poincaré biextension, this is the usual Weil pairing. The preceding construction shows that it is not an additional structure but the image, under the formalism of derived tensor products, of the universal biextension class.

3. Biextensions of group schemes (P, Q) by $\mathbb{G}_m$: generalities

3.1. The basic example

Recall first the most important example. If A is an abelian scheme over S , and A' is its dual, then the Poincaré invertible sheaf on $A \times_S A'$, rigidified along the two zero sections, defines a biextension of (A, A') by $\mathbb{G}_m$. Its universal property identifies A' with the functor which classifies extensions of

A by $\mathbb{G}_m$. Thus the Poincaré biextension is the geometric representative of the identity morphism

$$A' \longrightarrow A'.$$

The partial extension structures are the theorem of the square in each variable. Their compatibility is the theorem of the cube. This example is the model for the general discussion.

3.2. The case in which one variable is an abelian scheme

Suppose Q is an abelian scheme. For a commutative group scheme P , a biextension of (P, Q) by $\mathbb{G}_m$ may be viewed, by fixing the first variable, as a morphism from P to the sheaf of extensions of Q by $\mathbb{G}_m$. Since this sheaf is represented by the dual abelian scheme Q' , one obtains a description

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q')$$

under the hypotheses which ensure the usual representability and rigidity. In the reverse direction, a morphism $P \rightarrow Q'$ pulls the Poincaré biextension on $Q \times Q'$ back to a biextension of (P, Q) .

This construction is compatible with base change, since the dual abelian scheme and the Poincaré biextension commute with base change. It is also compatible with the pairings on torsion defined above.

3.3. The case in which one variable is finite locally free

Assume now that Q is finite locally free over S . A biextension of (P, Q) by $\mathbb{G}_m$ is equivalently an extension of P by the Cartier dual $Q^\vee$, under the standard hypotheses. Indeed, restricting the biextension to $P \times \{q\}$ for sections q of Q gives a system of extensions of P by $\mathbb{G}_m$, additive in q . By Cartier duality this is the same as an extension by $Q^\vee$.

Similarly, if P is finite locally free, biextensions of (P, Q) by $\mathbb{G}_m$ correspond to extensions of Q by $P^\vee$. The two descriptions are identified by the symmetry of biextensions and the symmetry of Cartier duality.

This gives a useful bridge between the theory of biextensions and the duality theory of finite flat group schemes. In particular, for finite flat n -torsion group schemes the pairing of Section 2 is exactly the pairing obtained from the corresponding Cartier-dual extension.

3.4. The unipotent case

If one of P or Q is unipotent and the other is of the type considered above, many biextension groups vanish. The reason is that homomorphisms from a unipotent group to a torus, or from an

abelian variety to an affine unipotent group, vanish under the usual hypotheses. Consequently, the Ext descriptions obtained above collapse to zero in the relevant degrees.

These vanishings will be used in reduction arguments: after decomposing a smooth connected group scheme into its abelian, toric, and unipotent parts, only the abelian and toric components contribute to biextensions by $\mathbb{G}_m$ in the situations needed later.

3.5. Summary of the preceding cases

The preceding discussion may be summarized as follows. If P and Q are group schemes over S belonging locally to the standard classes of abelian schemes, tori, finite locally free groups, and unipotent groups, then $\mathrm{Biext}^1(P, Q; \mathbb{G}_m)$ is computed by the corresponding duality objects: dual abelian schemes, character groups of tori, and Cartier duals of finite flat groups. The comparison is functorial and compatible with the pairings on torsion.

3.6. Two abelian schemes

Let P and Q be abelian schemes over S , and let Q' be the dual of Q . The preceding construction gives a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q').$$

By symmetry, this is also canonically

$$\mathrm{Hom}(Q, P').$$

The Poincaré biextension corresponds to the identity in the case $P = Q'$, and pullback of the Poincaré biextension gives the inverse map.

In particular, a biextension of (P, Q) by $\mathbb{G}_m$ determines, and is determined by, the bilinear correspondence between P and Q expressed by the associated homomorphism to the dual abelian scheme.

3.7. Semi-abelian variants

Suppose P is an extension of an abelian scheme by a torus, and Q is of a similar type. The computation of $\mathrm{Biext}^1(P, Q; \mathbb{G}_m)$ is obtained by devissage from the abelian and toric parts. The abelian–abelian part is governed by the dual abelian scheme; the toric part is governed by character groups; and the mixed parts are the evident homomorphism groups between the corresponding lattices and dual abelian schemes.

This form of the result is the one needed for Néron models. It is stable under smooth base change

and local for the fppf topology.

3.8. Principal consequences

For smooth group schemes over S whose geometric fibres are built from abelian varieties, tori, and additive groups, the category of biextensions by $\mathbb{G}_m$ is controlled by the duality data of those pieces. In particular, if one of the relevant Hom or Ext groups vanishes for structural reasons, the biextension is uniquely trivial. If an abelian part remains, the biextension is the pullback of the Poincaré biextension by the corresponding morphism to the dual.

4. Biextensions and the theorem of the cube

4.1. The cube condition

Let P, Q, G be sheaves of commutative groups on a topos. A biextension of (P, Q) by G can be characterized as a G -torsor on $P \times Q$, rigidified along the two axes, for which the two partial difference torsors satisfy the cocycle conditions in both variables. This is an abstract form of the theorem of the cube.

For group schemes, especially for abelian schemes, the theorem of the cube says that an invertible sheaf on a product which is rigidified on the coordinate axes and satisfies the usual compatibility on triples carries a unique biextension structure. Thus biextensions by $\mathbb{G}_m$ may often be recognized purely in terms of rigidified line bundles.

4.2. Smooth connected groups

If P and Q are smooth connected group schemes over a base S , the cube condition is checked fibrewise and then descended. The connectedness ensures that a morphism to a discrete group is determined by its value on the identity component, hence often vanishes. This is the mechanism behind the full faithfulness results used later.

4.3. Abelian varieties over a field

Let P and Q be abelian varieties over a field k . A rigidified line bundle on $P \times Q$ satisfying the cube condition defines a biextension of (P, Q) by $\mathbb{G}_m$. Conversely, every such biextension has this form. The theorem of the cube gives the necessary associativity and compatibility of the two partial group laws.

For abelian varieties this recovers the classical description:

$$\mathrm{Biext}^1(P, Q; \mathbb{G}_m) \simeq \mathrm{Hom}(P, Q').$$

4.4. On the automatic nature of the cube condition

It is natural to ask whether the cube condition imposed above follows automatically from the two rigidifications and from weaker additivity conditions. The answer depends on the class of group schemes. For abelian varieties the theorem of the cube gives the desired uniqueness. In more general situations, especially when disconnected fibres or affine parts intervene, one must retain the cube condition explicitly or verify it by devissage.

4.5–4.8. Devissage

The computations for smooth connected groups are obtained by filtering the group schemes by affine, toric, unipotent, and abelian parts. The biextension functor is exact enough in each variable to allow passage from the pieces to the whole. Vanishing for unipotent pieces and the standard duality for tori and abelian schemes give the required formulas.

4.9–4.13. Invertible sheaves and ampleness

Let P be a smooth connected group scheme over k , extension of an abelian variety by a torus or by a unipotent group. Rigidified invertible sheaves on P are controlled by the corresponding Picard functor. If an invertible sheaf $\mathcal{L}$ on P is ample, the associated morphism to the dual is an isogeny on the abelian quotient. The biextension attached to $\mathcal{L}$ is obtained by the usual formula

$$m^* \mathcal{L} \otimes p_1^* \mathcal{L}^{-1} \otimes p_2^* \mathcal{L}^{-1},$$

where m is the group law. The cube theorem identifies the compatibility needed for this line bundle to define a biextension.

5. Extensions and biextensions by constant truncated group schemes

Let S be a scheme, $T \subset S$ a closed subscheme, and Z a set or a group. Denote by Z_S the corresponding constant sheaf or constant group scheme on S , and by Z_T its restriction to T .

When Z is a group, the sheaf Z_S is an étale group scheme. The basic computations of Hom , Ext , and Biext with coefficients in such constant groups reduce to computations on connected

components. If P is proper over S , then morphisms $P \rightarrow Z_S$ are locally constant and hence factor through the component group. For an abelian scheme P , this gives

$$\mathrm{Hom}(P, Z_S) = 0$$

when Z is torsion-free and the fibres of P are connected.

More generally, if P is an abelian scheme over S , the Ext groups $\mathrm{Ext}^i(P, Z_S)$ are controlled by the component data on the boundary. Analogous formulas hold for Tor_i , and all these constructions commute with base change. These facts will be applied with $Z = \mathbb{Z}$, supported on a closed subscheme, in the study of the Néron model of $\mathbb{G}_m$.

Proposition 0.3. *Let S be a scheme, $T \subset S$ a closed subscheme, Z a torsion-free constant group, and P, Q abelian schemes over S . Under the standard connectedness hypotheses, the groups of extensions and biextensions by the constant group supported on T are described by the corresponding Hom groups from the component groups of the special fibres to Z , and by their derived functors.*

The proposition is a direct consequence of the exact sequences for Ext and Biext and the fact that proper connected fibres admit no non-constant maps to a discrete group.

6. Extensions and biextensions by the Néron model of $\mathbb{G}_m$

Let S be a scheme, $T \subset S$ a closed subscheme, and $U = S - T$. Let $\mathcal{N}$ be the Néron–Raynaud model of $\mathbb{G}_m$ relative to U . It fits, under the hypotheses used here, into an exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathcal{N} \longrightarrow i_*\mathbb{Z}_T \longrightarrow 0,$$

where $i : T \hookrightarrow S$.

For a commutative group scheme P over S , the extensions of P by $\mathcal{N}$ are easier to control than extensions by $\mathbb{G}_m$, because $\mathcal{N}$ has the Néron mapping property. If P is an abelian scheme, one gets

$$\mathrm{Ext}^1(P, \mathcal{N}) \simeq \mathrm{Ext}^1(P_U, \mathbb{G}_m)$$

by restriction to U . Similarly, for two abelian schemes P, Q ,

$$\mathrm{Biext}^1(P, Q; \mathcal{N}) \simeq \mathrm{Biext}^1(P_U, Q_U; \mathbb{G}_m).$$

The obstruction to replacing $\mathcal{N}$ by $\mathbb{G}_m$ lies in the corresponding groups with coefficients in $i_*\mathbb{Z}_T$.

If L is an invertible sheaf on $P \times Q$, rigidified along the axes, its associated biextension by $\mathbb{G}_m$ extends to a biextension by $\mathcal{N}$ when restricted to the open U . The failure of this $\mathcal{N}$ -biextension to

come from a $\mathbb{G}_m$ -biextension is measured by the component group term $i_*\mathbb{Z}_T$.

Proposition 0.4. *Let S be regular and let T be a divisor with suitable normal-crossing or unibranch hypotheses. Then the quotient sheaf*

$$j_*(\mathbb{G}_m)_U/(\mathbb{G}_m)_S$$

is represented, étale locally, by the constant group supported on T , and its formation is compatible with smooth base change.

In the case where T is geometrically unibranch, this quotient is simply

$$i_*\mathbb{Z}_T.$$

Consequently the exact sequence above gives explicit obstruction groups for the extension of $\mathbb{G}_m$ -extensions and $\mathbb{G}_m$ -biextensions.

Theorem 0.5. *Let S be regular, let T be a divisor of the preceding type, and let P, Q be abelian schemes over S . The obstruction to extending a biextension of (P_U, Q_U) by $\mathbb{G}_m$ to a biextension of (P, Q) by $\mathbb{G}_m$ is the image of its canonical extension by $\mathcal{N}$ in*

$$\mathrm{Biext}^1(P, Q; i_*\mathbb{Z}_T).$$

The corollary used later is the corresponding criterion for vanishing of this obstruction in terms of component groups.

7. Canonical prolongations of extensions and biextensions by $\mathbb{G}_m$

Let $S = \mathrm{Spec} V$ be a trait, with generic point η , closed point s , and residue field k . Let P be a smooth commutative group scheme over S , and write

$$\Phi = P_s/P_s^0$$

for the group of connected components of the special fibre.

Theorem 0.6. *Assume Φ is a torsion group, for instance P is of finite type over S . Then the restriction functor from extensions of P by $\mathbb{G}_m$ over S to extensions of P_η by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$, then to every extension E_η of P_η by $\mathbb{G}_m$ there is associated a canonical homomorphism*

$$d(E_\eta) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k$$

whose vanishing is necessary and sufficient for E_η to be the generic fibre of an extension of P by $\mathbb{G}_m$. In particular, if P_s is connected, restriction is an equivalence.

For biextensions the statement is analogous. Let P, Q be smooth commutative group schemes over S , and put

$$\Phi = P_s/P_s^0, \quad \Psi = Q_s/Q_s^0.$$

Theorem 0.7. *Assume that Φ or Ψ is torsion. Then restriction from biextensions of (P, Q) by $\mathbb{G}_m$ over S to biextensions of (P_η, Q_η) by $\mathbb{G}_m$ over η is fully faithful. If $n\Phi = 0$ or $n\Psi = 0$, then every generic biextension E_η has a canonical obstruction pairing*

$$d(E_\eta) : \Phi \otimes \Psi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k,$$

and this pairing vanishes if and only if E_η extends to a biextension of (P, Q) by $\mathbb{G}_m$ over S . If P_s or Q_s is connected, restriction is an equivalence.

The proof uses the Néron–Raynaud model $\mathcal{N}$ of $(\mathbb{G}_m)_\eta$. Restriction for biextensions by $\mathcal{N}$ is an equivalence by the Néron mapping property. Thus the question is reduced to passing from $\mathcal{N}$ to $\mathbb{G}_m$, using the exact sequence

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathcal{N} \longrightarrow \mathbb{Z}_s \longrightarrow 0.$$

For extensions, full faithfulness is equivalent to

$$\mathrm{Hom}(P, \mathbb{Z}_s) = 0,$$

which identifies with $\mathrm{Hom}(\Phi, \mathbb{Z})$ and therefore vanishes when Φ is torsion. The obstruction group is

$$\mathrm{Ext}^1(P, \mathbb{Z}_s) \simeq \mathrm{Hom}(\Phi, \mathbb{Q}/\mathbb{Z}).$$

For biextensions the same argument, with the exact sequence of Biext's, gives the obstruction group

$$\mathrm{Biext}^1(P, Q; \mathbb{Z}_s) \simeq \mathrm{Hom}(\Phi \otimes \Psi, \mathbb{Q}/\mathbb{Z}).$$

7.3. Complements

In the situation of the extension theorem, with $n\Phi = 0$, the obstruction $d(E_\eta)$ admits a torsion interpretation. The restriction of E_η to ${}_nP_\eta$ gives an extension of ${}_nP_\eta$ by μ_n . Passing to component groups gives

$${}_n\Phi \longrightarrow (\mathbb{Z}/n\mathbb{Z})_k,$$

which is the restriction of $d(E_\eta)$.

In the biextension case, a generic biextension E_η defines the torsion pairings discussed in Section 2.

These recover the obstruction pairing

$$d(E_\eta) : \Phi \otimes \Psi \rightarrow (\mathbb{Q}/\mathbb{Z})_k.$$

If $q \in Q(S)$ is a section and q_s its image in $\Psi(k)$, then the extension of P_η obtained from E_η by restriction along q_η has obstruction

$$d_q(x) = d(E_\eta)(x, q_s).$$

Thus the biextension obstruction is characterized by the extension obstructions obtained by fixing one variable.

The obstruction is symmetric under transposition. If ${}^t E_\eta$ denotes the transposed biextension of (Q_η, P_η) , then the diagram

$$\begin{array}{ccc} \Phi \otimes \Psi & \xrightarrow{d(E_\eta)} & (\mathbb{Q}/\mathbb{Z})_k \\ \downarrow & & \parallel \\ \Psi \otimes \Phi & \xrightarrow{d({}^t E_\eta)} & (\mathbb{Q}/\mathbb{Z})_k \end{array}$$

commutes. In particular, if $P = Q$ and E_η is symmetric, then the obstruction form on Φ is symmetric.

Finally, let $S' \rightarrow S$ be a dominant local morphism of traits and let

$$e(S'/S)$$

be the ramification index. Under base change one has

$$d(E_{\eta'}) = e(S'/S) d(E_\eta) \otimes_k k'.$$

When $e(S'/S) = 1$, for example for étale extensions, henselizations, strict henselizations, or completions, formation of the obstruction commutes with base change. In general, after a suitable finite extension of traits, the obstruction is killed and the extension or biextension prolongs.

Proposition 0.8. *Let S be reduced. If P is a smooth group scheme over S whose fibres at the maximal points are connected, then the functor from extensions of P by $\mathbb{G}_m$ to rigidified $\mathbb{G}_m$ -torsors on P is fully faithful. If S is normal and P has connected fibres, a rigidified torsor lies in the essential image if and only if this is true over every maximal point of S .*

The analogous statement holds for biextensions of (P, Q) : the functor to bitrivialized $\mathbb{G}_m$ -torsors on $P \times_S Q$, rigidified along the two unit sections, is fully faithful under the same connectedness assumptions, and over a normal base its essential image is detected at the maximal points.

Corollary 0.9. *Let S be normal and let P, Q be smooth group schemes over S , with connected*

maximal fibres and with one of P, Q having connected fibres. Assume that over every maximal point one of P_η, Q_η is an extension of an abelian variety by a torus, or more generally has a composition series whose factors are abelian varieties, tori, or additive groups. Then every bitrivialized torsor of the preceding kind comes from a unique biextension.

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Exposé IX. Néron models of abelian schemes and monodromy

0. Introduction

Let S be a trait with generic point $\eta = \operatorname{Spec} K$, closed point s , and residue field k . Let A_K be an abelian variety over K , and let A be its Néron model over S . The object of this exposé is to express the structure of the special fibre A_s , and in particular its group of connected components, in terms of the action of inertia on the Tate module of A_K .

The dual abelian variety is denoted A'_K , and its Néron model by A' . The Poincaré biextension of (A_K, A'_K) by $\mathbb{G}_m$ has a canonical prolongation to a biextension

$$W$$

of (A^0, A'^0) by $\mathbb{G}_m$. From it one obtains a canonical pairing of component groups

$$\Phi \times \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_k,$$

where

$$\Phi = A_s/A_s^0, \quad \Phi' = A'_s/A'^0_s.$$

This pairing will be related to the monodromy pairing on Tate modules.

Fix a prime ℓ invertible on S . The ℓ -adic Tate module $T_\ell(A_K(\overline{K}))$ is equipped with an action of

$$\pi = \operatorname{Gal}(\overline{K}/K),$$

and in particular of the inertia group I . The main constructions below single out two canonical submodules: the fixed part and the toric part. The orthogonality theorem identifies the toric part as the orthogonal complement of the fixed part for the Weil pairing attached to the canonical biextension.

1. Néron model of an abelian scheme

1.0. Notation and the canonical pairing

Let A_K be an abelian variety over K , A'_K its dual, and A, A' their Néron models over S . Write

$$\Phi = A_s/A_s^0, \quad \Phi' = A'_s/A'^0_s.$$

The canonical biextension W_K of (A_K, A'_K) by $\mathbb{G}_m$ extends canonically to a biextension

$$W$$

of (A^0, A'^0) by $\mathbb{G}_m$. Applying the obstruction formalism of Exposé VIII to this biextension gives a pairing

$$\Phi \times \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_k.$$

This is the component pairing associated with the Poincaré biextension. It is functorial in A_K and compatible with duality.

The restriction of W to the generic fibre is W_K . The uniqueness part of the canonical prolongation theorem ensures that W is uniquely determined by W_K , and hence that the pairing on component groups is canonical.

1.1. Non-degeneracy properties

The pairing

$$\Phi \times \Phi' \rightarrow (\mathbb{Q}/\mathbb{Z})_k$$

is expected, and in the situations used below is known, to be non-degenerate in the appropriate sense. More precisely, after restricting to the finite component groups which arise for Néron models of abelian varieties over a trait, it induces homomorphisms

$$\Phi \longrightarrow \mathrm{Hom}(\Phi', \mathbb{Q}/\mathbb{Z}), \quad \Phi' \longrightarrow \mathrm{Hom}(\Phi, \mathbb{Q}/\mathbb{Z}).$$

Raynaud's results identify these maps as isomorphisms under the standard hypotheses. This is one of the principal inputs in the orthogonality theorem.

1.2. Behaviour under isogenies

Let C_K be another abelian variety over K , with Néron model C , and let

$$u_K : C_K \rightarrow A_K$$

be a homomorphism. By the Néron mapping property it extends uniquely to

$$u : C \rightarrow A.$$

If u_K is an isogeny, then u induces isogenies on the connected components of the special fibres and on the corresponding toric and abelian parts. In particular the numerical invariants attached to

the reduction of A_K are invariant under isogeny. The same applies to A_K and its dual.

1.3. The invariants λ, μ, α

Let A_s^0 be the identity component of the special fibre. Its Chevalley decomposition has an affine part whose maximal torus has rank μ , an abelian quotient of dimension α , and a unipotent part of rank λ . These integers satisfy

$$\dim A_K = \lambda + \mu + \alpha.$$

The dual abelian variety has the same invariants. This follows from the isogeny-invariance just mentioned, together with the duality between the toric and component data.

1.4. The canonical biextension on the identity components

The canonical biextension W on (A^0, A'^0) restricts on the special fibre to a biextension

$$W_s$$

of $(A_s^0, A_s'^0)$ by $\mathbb{G}_m$. This biextension is the object through which the ℓ -adic pairings on the fixed parts of the Tate modules are compared with the geometry of the special fibre. Its non-degeneracy on the abelian quotients and its vanishing on toric directions are the essential facts used below.

2. Fixed and toric parts of $T_\ell(A)$. Criteria for good reduction

2.1. The Tate module and inertia

Let ℓ be a prime distinct from the residue characteristic p . The Tate module

$$T_\ell(A_K) = T_\ell(A_K(\overline{K}))$$

is a free $\mathbb{Z}_\ell$ -module of rank $2 \dim A_K$, equipped with a continuous action of $\pi = \text{Gal}(\overline{K}/K)$. Let $I \subset \pi$ be the inertia group.

If S is henselian, the finite étale group schemes ${}_{\ell^\nu}A$ over S identify their sections over the maximal unramified extension with the inertia invariants in ${}_{\ell^\nu}A_K(\overline{K})$. Passing to the inverse limit gives the fixed part:

$$T_\ell(A_K)^f = T_\ell(A_K)^I.$$

2.2. Definition of the fixed part

The fixed part is the ℓ -adic sheaf on S whose generic fibre is the inertia-invariant submodule of $T_\ell(A_K)$. Equivalently, when S is henselian, it is

$$T_\ell(A_K)^f = \varprojlim_{\nu} \ell^\nu A(K^{\text{nr}}),$$

where K^{nr} is the maximal unramified extension of K in $\overline{K}$.

The specialization map identifies this module with the Tate module of the identity component of the special fibre:

$$T_\ell(A_K)^f \simeq T_\ell(A_s^0)$$

after the standard passage through the Néron model. More precisely, the fixed part is characterized as the lisse $\mathbb{Z}_\ell$ -sheaf whose special fibre is $T_\ell(A_s^0)$.

Proposition 0.10. *Assume S henselian and $\ell \neq p$. The submodule of $T_\ell(A_K(\overline{K}))$ corresponding to the fixed part is the submodule of invariants:*

$$T_\ell(A_K)^f = T_\ell(A_K(\overline{K}))^I.$$

For every integer $m > 0$ prime to p one has

$${}_m A(K^{\text{nr}}) = ({}_m A(\overline{K}))^I.$$

This is because ${}_m A$ is finite étale over S . Passing to the inverse limit over $m = \ell^\nu$ proves the proposition.

2.2.6. Isogenies and the fixed part

Let C_K be another abelian variety with Néron model C , and let $u_K : C_K \rightarrow A_K$ be an isogeny. Then

$$T_\ell(u_K) : T_\ell(C_K) \rightarrow T_\ell(A_K)$$

induces an isogeny on fixed parts

$$T_\ell(C_K)^f \longrightarrow T_\ell(A_K)^f.$$

The induced morphism on the identity components of special fibres gives an isogeny

$$T_\ell(C_s^0) \longrightarrow T_\ell(A_s^0).$$

It also induces an isogeny on maximal tori and, after a harmless extension of the residue field, on the abelian quotients of the identity components.

Consequently the invariants

$$\lambda, \quad \mu, \quad \alpha$$

of the reduction are invariant under isogeny. In particular A_K and its dual A'_K have the same invariants.

2.2.9. Good reduction

Let A_K be an abelian variety over K , A its Néron model, and $\ell \neq p$. The following conditions are equivalent.

- (i) There exists an abelian scheme over S with generic fibre A_K .
- (ii) The Néron model A is an abelian scheme.
- (iii) The identity component A^0 is an abelian scheme.
- (iv) The special fibre A_s is an abelian variety.
- (v) The identity component A_s^0 is an abelian variety, equivalently $\lambda = \mu = 0$.
- (vi) The Néron model A is proper over S .
- (vii) The inertia group I acts trivially on $T_\ell(A_K(\overline{K}))$.
- (viii) Equivalently, for every ν , I acts trivially on ${}_{\ell^\nu}A_K(\overline{K})$.

This is the Néron–Ogg–Shafarevich criterion for good reduction. An abelian scheme satisfying (i) is the Néron model of its generic fibre and is therefore unique up to unique isomorphism.

The proof uses the Néron mapping property for abelian schemes over a regular base, the structure of A_s^0 , and the equality between the fixed part of the Tate module and the inertia invariants. Trivial inertia action forces the fixed part to be the whole Tate module, which is equivalent to the absence of toric and unipotent contribution in the special fibre.

2.3. The toric part

Assume again that S is henselian and $\ell \neq p$. Let T_0 be the maximal torus of A_s^0 . The inclusion

$$T_\ell(T_0) \subset T_\ell(A_s^0)$$

defines a canonical lisse $\mathbb{Z}_\ell$ -subsheaf of $T_\ell(A^0)$, called the toric part:

$$T_\ell(A^0)^t \subset T_\ell(A^0) \subset T_\ell(A).$$

Its special fibre is

$$T_\ell(A^0)^t \times_S s = T_\ell(T_0).$$

On the generic fibre this gives a π -stable submodule

$$T_\ell(A_K)^t \subset T_\ell(A_K(\overline{K})),$$

the toric submodule of the Tate module. Thus

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K)$$

is a canonical three-step filtration. The same construction applies to the dual abelian variety.

2.4. The orthogonality theorem

Let A'_K be the dual abelian variety. The Poincaré biextension gives the Weil pairing

$$e_\ell : T_\ell(A_K) \times T_\ell(A'_K) \longrightarrow \mathbb{Z}_\ell(1).$$

Theorem 0.11. *Assume S henselian and $\ell \neq p$. With respect to the pairing e_ℓ , the toric part of $T_\ell(A_K)$ is the orthogonal complement of the fixed part of $T_\ell(A'_K)$ inside the fixed part of $T_\ell(A_K)$:*

$$T_\ell(A_K)^t = T_\ell(A_K)^f \cap (T_\ell(A'_K)^f)^\perp.$$

The symmetric statement with A and A' interchanged also holds.

The first inclusion follows by extending the Poincaré biextension to (A^0, A'^0) . On special fibres, it is enough to prove that the pairing

$$T_\ell(T_0) \times T_\ell(A_s'^0) \rightarrow \mathbb{Z}_\ell(1)$$

is zero; this is a consequence of the vanishing of biextensions between a torus and the relevant connected part. For the reverse inclusion, one reduces to the special fibre and uses the non-degeneracy, due to Raynaud, of the biextension W_s on the abelian quotient.

2.5. Polarized form

If ξ is a polarization of A_K , it identifies $T_\ell(A_K)$ with a submodule of $T_\ell(A'_K)$ up to isogeny and gives an alternating form

$$e_\ell^\xi : T_\ell(A_K) \times T_\ell(A_K) \rightarrow \mathbb{Z}_\ell(1).$$

Let

$$V = T_\ell(A_K)^f, \quad W = T_\ell(A_K)^t.$$

The orthogonality theorem becomes the simple formula

$$V^\perp = W.$$

The successive ranks in the induced filtration recover the numerical invariants λ, μ, α of the special fibre.

3. The case of semi-stable reduction

The semi-stable case is characterized by the absence of unipotent part in the identity component of the special fibre. Equivalently, the identity component is an extension of an abelian variety by a torus. The Tate module filtration then has only the fixed and toric contributions described above.

We first record a general result in the style of SGA 3. Let G be a commutative S -group scheme, flat, separated, and of finite presentation, let H be a quasi-separated group scheme locally of finite type over S , and let

$$u : G \rightarrow H$$

be a homomorphism with kernel K . For $s \in S$, and for n prime to the residue characteristic, the group scheme ${}_nK$ is étale of finite type near s , and its geometric ranks specialize divisibly:

$$\text{rank}({}_nK_{\bar{s}}) \mid \text{rank}({}_nK_{\bar{t}}).$$

This gives inequalities among the toric, abelian, and unipotent ranks of the fibres. In particular, if the generic connected kernel is unipotent, then so is the special connected kernel. If G_s has no unipotent part, H is smooth, and u_s is an isogeny, then u is flat and quasi-finite near s .

These facts are used to prove the semi-stable reduction theorem: after a finite separable extension of K , an abelian variety acquires semi-stable reduction. In the language of Tate modules this is reflected in the quasi-unipotence of the inertia action and in the existence, after finite extension, of the canonical filtration whose graded pieces are the toric and abelian parts.

Further continuation of Exposé IX

The following paragraphs continue the discussion of semistable reduction, conductors, and the filtrations on Tate modules. **4.4.** Place ourselves under the conditions of 4.3 b). Then for every element g in π' whose image in π_0 is of the form f_q^n with $n \geq 0$, the eigenvalues of the induced automorphisms of $U^{\text{ét}}$, $U^{\text{ét}}/U^{\text{ét}}$, $U/U^{\text{ét}}$ are algebraic integers whose absolute values are respectively q^n , $q^{n/2}$ and 1. Concerning the two extreme pieces, one can say, more precisely, that the corresponding eigenvalues are respectively $(q^n \varepsilon_i)_{1 \leq i \leq \mu}$ and $(\varepsilon_i)_{1 \leq i \leq \mu}$, where the ε_i ($1 \leq i \leq \mu$) are roots of unity.

Indeed, formula (4.1.2) gives for the last piece the eigenvalues $q^n/(q^n \varepsilon_i) = \varepsilon_i^{-1}$, but as ε_i is a root of unity, we have $\varepsilon_i^{-1} = \overline{\varepsilon_i}$ (complex conjugate), and as the family of ε_i is stable under complex conjugation (as the family of solutions of an equation with integer coefficients), the family of ε_i^{-1} is also that of the ε_i , which proves the last assertion of 4.4.

4.5. We shall give an application of 4.3 a) to the theory of local conductor of abelian schemes. Suppose k algebraically closed (*). Recall (SGA 5 X 6.2 or [20]) that for every commutative étale ℓ -group scheme F_K over K , ($\ell \neq p$), one defines an integer

$$a_s(F_K) = a_s(i_*(F_K)) \quad (1)$$

($i: \eta \rightarrow S$ denoting the inclusion), in the following way : one chooses a finite Galois extension K' of K , with group Γ , which splits F_K , so that F_K is defined by a finite ℓ -group M on which Γ acts. Moreover, one associates to the extension K' of K a certain *projective* $\mathbb{Z}_\ell[\Gamma]$ -module (defined up to isomorphism)

$$\text{Sw} = \text{Sw}(K'/K, \mathbb{Z}_\ell) , \quad (2)$$

and one sets

$$a_s(F_K) = \text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(\text{Sw}, M) . \quad (3)$$

One verifies (loc. cit.) that this expression does not depend on the choice of the extension K'/K . Besides, the fact that Sw is a projective $\mathbb{Z}_\ell[\Gamma]$ -module implies that $a_s(F_K)$ behaves additively for exact sequences.

We shall apply this definition to the case of the ℓ -adic sheaf $\mathcal{A}_K$, where $\mathcal{A}_K$ is our abelian scheme over K . One has thus, in this case, an integer

$$a_s(\mathcal{A}_K; \ell) = \text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(\text{Sw}, U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell) , \quad U_\ell = T_\ell(A(\overline{K})) , \quad (4)$$

Γ being chosen as was said (choice depending a priori on ℓ). The integer (4.5.4) is called the *wild conductor exponent* (or “sauvage”) (*) of the abelian scheme $\mathcal{A}_K$ with respect to the discrete

valuation ring $V \subset K$; by passing to the henselisation, it is of course defined without supposing V henselian, and it is immediately reduced to the case where S is *strictly local*. We propose to show the

Corollary 0.12 (4.6). *The wild conductor exponent (4.5.4) does not depend on the choice of the prime number $\ell \neq p$.*

We may assume S strictly local. We shall give the calculation of (4.5.4), replacing Sw by any other projective $\mathbb{Z}_\ell[\Gamma]$ -module L . Note that the filtration (4.1.1) of U_ℓ is formed by $\mathbb{Z}_\ell$ -submodules which are manifestly direct factors, so this filtration induces an analogous filtration of $U(\ell) = U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell$, whose successive factors are deduced from the factors V_i ($i = 1, 2, 3$)

$$U^{\text{ét}}, \quad U^{\text{ét}}/U^{\text{ét}}, \quad U/U^{\text{ét}}$$

appearing in the filtration (4.1.1) by tensorisation with $\mathbb{F}_\ell$. Of course, this filtration is stable under Γ , and gives by additivity a decomposition of $\text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell))$ into a sum of three terms, lengths of the groups

$$\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell) \simeq (\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i)) \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell, \quad (5)$$

the isomorphism written in (*) resulting again from the projectivity hypothesis made on L ; one only assumes K' chosen large enough for Γ to operate on the V_i . Now $\text{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i)$ is evidently a torsion-free $\mathbb{Z}_\ell$ -module, hence its *rank* is equal to the right hand side of (*). If φ_i is the character of the representation of Γ in V_i , and ψ the character of the representation of Γ in $\check{L}$, one therefore finds

$$\begin{aligned} \text{length Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell)) &= \sum_{1 \leq i \leq 3} N^{-1} \sum_{g \in \Gamma} \psi(g) \varphi_i(g) \\ &= N^{-1} \sum_{g \in \Gamma} \psi(g) \varphi(g), \quad \text{with } N = \text{card } \Gamma, \end{aligned} \quad (6)$$

where $\varphi(g) = \sum \varphi_i(g)$ is also the function on Γ deduced by passage to the quotient from the function on the inertia group I , character of the monodromy representation of I on U . Now by virtue of 4.3 a) this character is a function with *rational integer values independent of ℓ* . Taking $L = \text{Sw}$, it is the same for the character $\psi(g)$ of L , which is equal to $a - N\delta + 1$, where δ is the “Dirac function” of Γ , and a the “Artin conductor” of K'/K [26]. This therefore proves 4.6, while exhibiting at the same time the following expression, independent of ℓ , for the wild conductor exponent :

$$a_s(\mathcal{A}_K) = N^{-1} \sum_{g \in \Gamma} (a(g) - N\delta(g) + 1) \cdot \varphi(g), \quad (7)$$

where Γ is the Galois group of a finite Galois extension K' of K splitting $U^{\text{ét}}$ (for example the smallest such extension, cf. 4.1), $a(g)$ the Artin conductor on Γ , and $\varphi(g)$ the trace function on Γ

made explicit in (4.3.0). In the case of semi-stable reduction, one can take $K' = K$, hence $\Gamma = e$, and one finds

$$a_s(\mathcal{A}_K) = 0 \quad (\text{semi-stable reduction case}). \quad (8)$$

(*) It would in fact suffice that k be *perfect*, but by passing to the strict henselisation of S , one immediately reduces to the case under consideration.

4.7. To conclude, let us make explicit a criterion of semi-stable reduction which follows from the general reflections of 4.2 :

Proposition 0.13 (4.7). (M. RAYNAUD's criterion for semi-stable reduction). *Let S be a trait, A_K an abelian scheme defined over its function field K , n an integer ≥ 1 prime to the residual characteristic p (*), and such that ${}_n A_K$ is unramified with respect to S , i.e. such that the inertia group acts trivially on ${}_n A(\overline{K})$. Then, either A_K has semi-stable reduction over S , or $n = 2$, and if S is strictly local, A_K acquires semi-stable reduction over a finite Galois extension K' of K with Galois group isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$.*

Proof. We may obviously assume S strictly local. With the notation of 4.1 and 4.2, let K' be the smallest possible, i.e. assume that Γ acts *faithfully* on A_0^{*0} . By hypothesis, Γ acts trivially on ${}_n A(\overline{K})$, and a fortiori on ${}_n A(\overline{K})^{\text{ét}} \simeq {}_n A_0^*(k) \supset {}_n A_0^{*0}(k)$. We conclude by means of the two following lemmas :

Lemma 0.14 (4.7.1). (J.P. SERRE). *Let G be a commutative group scheme over a field k , extension of an abelian scheme by a torus, n an integer ≥ 2 . Then every automorphism of finite order of G inducing the identity on ${}_n G$ is reduced to the identity if $n \neq 2$, and is the identity or of order two if $n = 2$.*

Lemma 0.15 (4.7.2). *A finite group in which every element $\neq 1$ is of order 2 is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$, for a suitable $r \geq 0$.*

4.7.3. Let us recall the proof of 4.7.1, which is well known. One notes that thanks to the fact that $n \cdot \text{id}_G$ is an epimorphism, the hypothesis that u induce the identity on ${}_n G$, i.e. that $1 - u$ induce 0 on ${}_n G$, is expressed by the condition that one has $1 - u = nv$ in the ring of endomorphisms of G . To prove the announced conclusion, one is reduced (decomposing the finite cyclic group generated by u into its p -primary components, for the various prime numbers p dividing the order of u) to the case where u is of order p^N , a power of a prime number. Let E be the subring of $\text{End}(G)$ generated by v , which is a quotient of $\mathbb{Z}[T]/(T^{p^N} - 1)$, hence of finite type over $\mathbb{Z}$, and which moreover is torsion-free (this is a formal consequence of the fact that G is divisible). It thus embeds into $E_{\mathbb{Q}} = E \otimes_{\mathbb{Z}} \mathbb{Q}$, which appears as a quotient of $\mathbb{Q}[T]/(T^{p^N} - 1)$, which is a direct sum of finite field extensions of $\mathbb{Q}$. It is thus the same for $E_{\mathbb{Q}}$. Considering the images of E, u, v in the

component fields K_i of $E_{\mathbb{Q}}$, one is thus reduced to seeing that if $K = \mathbb{Q}(\mu)$ is a field generated by a primitive p^n -th root of unity μ , and if n is an integer ≥ 2 such that $\mu - 1 \equiv 0 \pmod{n}$ (in the ring of integers of K), then one has $\mu = 1$, or $\mu = -1$ and $n = 2$. Now this follows immediately from the well-known fact that if $\mu \neq 1$ i.e. $m \geq 1$, the ideal $(\mu - 1)$ is prime and the ideal (p) is a power thereof, whose exponent is > 1 except if $p = 2, m = 1$ i.e. $\mu = -1$. From the fact that $(\mu - 1)$ is prime and divides p it follows indeed that $n \geq 2$ can divide $\mu - 1$ only if $n = p$, and p divides $\mu - 1$ only if $(\mu - 1) = (p)$ i.e. $\mu = -1$, hence $n = 2$.

4.7.4. Let us prove 4.7.2. It suffices to prove that Γ is commutative. Now if $x, y \in \Gamma$, one will have $(xy)(xy) = 1$, which can also be written, since $x^2 = 1, y^2 = 1$,

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx, \quad \text{qed.}$$

□

Remark 0.16 (4.7.5). *The proof given of 4.7.3, together with 4.7.2, shows us that if G is a divisible commutative group of a topos, $n \geq 2$ an integer, and Γ a finite group, acting faithfully on G while inducing the identity on ${}_nG$, then Γ is reduced to the identity if $n \geq 3$, and is isomorphic to a group $(\mathbb{Z}/2\mathbb{Z})^r$ if $n = 2$.*

§ 5. The orthogonality theorem in the case $\ell = p$ (semi-stable case). Duality of abelian schemes B_0 and B'_0 . Characterisation of the fixed part. Criteria for good reduction, essentially good reduction, and semi-stable reduction.

5.1. Place ourselves in the case 2.2.3 where S is henselian and A^0 is ℓ -divisible, so that the *fixed part*

$$T_\ell(A^0)^f \hookrightarrow T_\ell(A)$$

is defined as the Barsotti–Tate pro- ℓ -group formed by the $(\ell^\nu A^0)^f$. We shall now give a definition of the toric part (2.3.2) of $T_\ell(A^0)$, independent of the hypothesis $\ell \neq p$ made in 2.3, but assuming on the other hand that S is *complete* (*). For this, introduce for every integer $n \geq 0$ the local artinian scheme

$$S_n = \operatorname{Spec}(V/\mathfrak{m}^{n+1}) , \quad (9)$$

n -th infinitesimal neighbourhood of s in S , and for every S -scheme X let $X_n = X \times_S S_n$. As $(A^0)_n = (A_n)^0$ is smooth over S_n , it is known (SGA 3 IX 3.6 bis) that T_0 is the special fibre of a unique subtorus

$$T_n \subset A_n^0 . \quad (10)$$

According to the uniqueness of the T_n , these are deduced from one another by the base changes $S_{n'} \rightarrow S_n$. In other terms, they define a *formal torus* $\hat{T}$ of the formal group $\hat{A}^0$ (in the category of formal schemes over S) formal completion of A^0 along the special fibre :

$$\hat{T} \subset \hat{A}^0 . \quad (11)$$

As the functor $X \mapsto \hat{X}$ (formal completion along the special fibre) induces an equivalence between the category of finite schemes over S , and that of formal finite schemes over S (EGA III 4.8), we shall simply identify a finite scheme over S with the corresponding formal scheme. With this identification, the definition of the “fixed parts” immediately gives us for every $\nu \geq 0$:

$$(\ell^\nu A^0)^f = \ell^\nu(\hat{A}^0) , \quad (12)$$

which can also be written as an identity of projective systems

$$T_\ell(A^0)^f = T_\ell(\hat{A}^0) . \quad (13)$$

One thus obtains a sub-pro-Barsotti–Tate group of the fixed part of $T_\ell(A^0)$ by taking $T_\ell(\hat{T}) \subset T_\ell(\hat{A}^0)$. We then define the *toric part* of $T_\ell(A^0)$ as

$$T_\ell(A^0)^t \stackrel{\text{dfn}}{=} T_\ell(\hat{T}) \subset T_\ell(\hat{A}^0) = T_\ell(A^0)^f . \quad (14)$$

So by definition, one will have

$$T_\ell(A^0)^t = (\ell^\nu \hat{T})_{\nu \geq 0} , \quad (15)$$

the finite and flat S -group $\ell^\nu \hat{T}$ itself being described as the inductive limit of the finite and flat S_n -groups

$$\ell^\nu(T_n) , \quad n \text{ variable} , \quad (16)$$

i.e. its affine algebra is described as the projective limit of the affine algebras of the groups (5.1.8). Of course, one will have

$$(T_\ell(A^0)^t)_n = T_\ell(T_n) , \quad (17)$$

which for $n = 0$ is none other than formula (2.3.3) by which we had characterised the toric part (as pro-finite-étale group) in the case $\ell \neq p$. This shows that the definition (5.1.6) is compatible with that given in 2.3 when $\ell \neq p$.

Passing to the generic fibres, the fixed and toric parts of $T_\ell(A^0)$ define fixed and toric parts in $T_\ell(A_K)$:

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K) . \quad (18)$$

These are inclusions of pro- ℓ -Barsotti–Tate groups. Consider the analogous filtration, corresponding to the dual abelian scheme A'_K :

$$T_\ell(A'_K)^t \subset T_\ell(A'_K)^f \subset T_\ell(A'_K) . \quad (19)$$

Theorem 0.17 (5.2). *Suppose that S is a complete trait, and let A_K be an abelian scheme over K , A'_K the dual abelian scheme, ℓ a prime number such that the connected Néron model A^0 is ℓ -divisible (cf. 2.2.1), so that we then dispose of the filtrations (5.1.10) and (5.1.11) of the pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$. This being posited, the toric parts are reconstructed in terms of the fixed parts by the formula*

$$T_\ell(A_K)^t = T_\ell(A_K)^f \cap (T_\ell(A'_K)^f)^\perp \quad (20)$$

and the analogous formula obtained by exchanging the roles of A_K and A'_K , where the sign $\perp$ denotes the orthogonal with respect to the canonical pairing (1.0.2).

Finally, when A_K has semi-stable reduction over S (which results from the other hypotheses in the case $\ell = p$), then one even has the formula

$$T_\ell(A_K)^t = (T_\ell(A'_K)^f)^\perp . \quad (21)$$

5.2.3. The meaning of formula (5.2.1) being quite clear if ℓ is prime to the characteristic of K (where it can be interpreted by passing to ordinary $\mathbb{Z}_\ell$ -modules, with action of $\pi = \text{Gal}(\overline{K}/K)$, formed of the $\overline{K}$ -points of the pro-groups in play, as in 2.4), let us make it explicit in the general

case, using the notion of dual

$$D(G) = \text{Hom}(G, T_\ell(\mathbb{G}_m)) \quad (22)$$

of a pro- ℓ -Barsotti–Tate group $G = (G(\nu))$, which by definition is the projective system of Cartier duals $D(G(\nu))$ (SGA 3 VII 3.3.1 and VIII 1), where the transition morphisms are defined in an evident way (by transposition from the morphisms of *inclusion* between the $G(\nu)$, for example). The duality theory of abelian varieties teaches us that the pairing mentioned in 5.2 in fact defines an *isomorphism*

$$T_\ell(A'_K) \xrightarrow{\sim} D(T_\ell(A_K)) , \quad (23)$$

whence by composition a morphism

$$T_\ell(A_K)^f \longrightarrow D(T_\ell(A'_K)^f) , \quad (24)$$

and formula (5.2.1) asserts that the kernel of the preceding homomorphism (which a priori is a sub-pro-group of the first term) is none other than the toric part $T_\ell(A_K)^t$. We shall note that *it is an isomorphism of pro-objects, and not of projective systems* : we do not assert (and it would be false in general) that for every integer ν , $T_\ell(A_K)^t(\nu)$ is equal to the kernel of $T_\ell(A_K)^f(\nu) \rightarrow D(T_\ell(A'_K)^f)(\nu)$; the latter could a priori be larger. (We shall see however below that it is not so if A_K has semi-stable reduction, which is in particular the case, according to the hypotheses made in 5.2, when $\ell = p$.)

5.2.4. The proof of (5.2.1) merely repeats, practically without change, that of 2.4. One notes that the induced pairing

$$T_\ell(A_K)^f \times T_\ell(A'_K)^f \longrightarrow T_\ell(\mathbb{G}_{mK}) \quad (25)$$

giving rise to (5.2.3.3) is induced by a pairing

$$T_\ell(A^0)^f \times T_\ell(A'^0)^f \longrightarrow T_\ell(\mathbb{G}_{mS}) \quad (26)$$

of Barsotti–Tate pro-groups over S , itself induced by the pairing of pro-groups

$$T_\ell(A^0) \times T_\ell(A'^0) \longrightarrow T_\ell(\mathbb{G}_{mS}) \quad (27)$$

deduced from the biextension W of (A^0, A'^0) by $\mathbb{G}_{mS}$. Hence, the pairing

$$T_\ell(A_0^0) \times T_\ell(A_0'^0) \longrightarrow T_\ell(\mathbb{G}_{mk}) , \quad (28)$$

deduced from (5.2.4.2) by passage to the special fibres, is none other than that deduced from the biextension W_0 of $(A_0^0, A_0'^0)$ by $\mathbb{G}_{mk}$. By virtue of VIII 4.11 and of 1.5, the left annihilator of the pairing (5.2.4.4) is therefore equal to $T_\ell(T_0)$, which is none other than the special fibre of the toric part $T_\ell(A^0)^t$. One thus concludes, returning to the definition of the left annihilators made explicit

in 5.2.3, by means of the

Lemma 0.18 (5.2.5). *Let $u: G \rightarrow H$ be a homomorphism of pro- ℓ -Barsotti–Tate groups over S . If $u_0: G_0 \rightarrow H_0$ is zero, then u is zero. If $u_0: G_0 \rightarrow H_0$ is an isogeny (resp. an isomorphism), u is an isogeny (resp. an isomorphism).*

We indeed apply the first assertion of 5.2.5 to the composed morphism

$$T_\ell(A^0)^t \longrightarrow T_\ell(A^0)^f \longrightarrow D(T_\ell(A^0)^f) ,$$

where the second morphism is that deduced from the pairing (5.2.4.2); one finds that this composition is zero, hence that the pairing under consideration vanishes on the toric part on the left. By symmetry, it therefore vanishes on the toric part on the right, hence it comes from a pairing on the quotients, or “abelian pieces”

$$T_\ell(A^0)^{\text{ab}} \times T_\ell(A'^0)^{\text{ab}} \longrightarrow T_\ell(\mathbb{G}_m) ,$$

whose quotients are again Barsotti–Tate pro-groups, as one easily verifies. It remains to prove that the morphism corresponding to this pairing

$$T_\ell(A^0)^{\text{ab}} \longrightarrow D(T_\ell(A'^0)^{\text{ab}})$$

is an isogeny, which again results from 5.2.5.

We shall leave to the reader the (easy) proof of 5.2.5, which is a result valid over every noetherian connected base. We shall find moreover below (7.4.7), by a somewhat more precise analysis of the biextensions W_n of the $(A_n^0, A_n'^0)$, a fairly obvious direct proof of the two particular cases where we had to use 5.2.5.

5.2.6. Suppose now that A_K has semi-stable reduction over S , so that we may take for ℓ any prime number (distinct or not from p). Moreover, by virtue of the Galois criterion for semi-stable reduction 3.5, one has for $\ell \neq p$

$$(T_\ell(A'_K)^f)^\perp (= T_\ell(A_K)^t) \subset T_\ell(A_K)^f . \quad (29)$$

It remains to prove, to finish the proof of 5.2, that this inclusion is valid without restriction on ℓ .

Indeed, the left hand side is canonically isomorphic to $D(T_\ell(A'_K)/T_\ell(A'_K)^f)$, and is thus of “rank” (or “height”) equal to $2n - 2\alpha - \mu = \mu$ (notation of (2.1.9), where by hypothesis $\lambda = 0$), which is also that of $T_\ell(A_K)^t$. As the inclusion

$$T_\ell(A_K)^t \subset T_\ell(A_K)$$

is *strict*, in the sense that it already induces a monomorphism for all terms of the projective systems under consideration, it follows easily that the first member of (*) is maximal among the sub-pro-Barsotti–Tate groups of $T_\ell(A_K)$ having the same rank as it, so that one indeed has the equality of (5.2.7), which is equivalent to the inclusion between the extreme terms of 5.2.7 by the orthogonality theorem.

5.3. Using the correspondence by orthogonality between strict sub-pro-Barsotti–Tate groups (in the precise sense explained above) of the dual pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$, one deduces formally from (5.2.2) that the pairing (1.0.2) between these pro-groups induces a pairing

$$T_\ell(A_K)^{\text{ab}} \times T_\ell(A'_K)^{\text{ab}} \longrightarrow T_\ell(\mathbb{G}_{mk}) \quad (30)$$

which is a *perfect duality*, i.e. identifying each of the two factors of the first member with the dual of the other (in the sense of 5.2.2). (NB the exponents ab denote the “abelian parts”, quotient of the fixed part by the toric part). It is clear moreover that (5.3.2) is induced by the pairing

$$T_\ell(A^0)^{\text{ab}} \times T_\ell(A'^0)^{\text{ab}} \longrightarrow T_\ell(\mathbb{G}_{mS}) \quad (31)$$

deduced by passage to the quotients from the pairing (5.2.4.2). This being a perfect duality on the generic fibres, i.e. the corresponding homomorphism

$$T_\ell(A^0)^{\text{ab}} \longrightarrow D(T_\ell(A'^0)^{\text{ab}}) \quad (32)$$

being an isomorphism on the generic fibres, we obviously want to deduce from this that it is the same over all of S , or equivalently, over the special fibres. This deduction is valid, trivially, if $\ell \neq p$; it is also valid, at least in the case of unequal characteristics, if $\ell = p$, thanks to a deep theorem of TATE [33, th. 4]. We shall however be able to establish this *directly*, without the help of Tate’s theorem, below (7.4.6).

To make precise the geometric meaning of the result we desire to establish, note that the pairing deduced from (5.3.3) by passage to the special fibres is none other than the pairing

$$T_\ell(B_0) \times T_\ell(B'_0) \longrightarrow T_\ell(\mathbb{G}_{mk}) \quad (33)$$

deduced from the biextension W_0 of (A_0^0, A'^0_0) by passage to the quotient, where B_0 and B'_0 denote respectively the abelian parts of A_0^0 and A'^0_0 . As, by virtue of VIII 4.8, the biextension W_0 comes from a biextension W_0^{ab} of (B_0, B'_0) by $\mathbb{G}_{mk}$, determined up to unique isomorphism, the pairing (5.3.5) can also be interpreted as deduced from the divisor correspondence W_0^{ab} , and the

corresponding homomorphism

$$T_\ell(B_0) \longrightarrow D(T_\ell(B'_0)) \xrightarrow{\sim} T_\ell(D(B'_0)) \quad (34)$$

(where in the last term, the letter D denotes the dual abelian scheme) is none other than that deduced by functoriality of T_ℓ from the homomorphism

$$B_0 \longrightarrow D(B'_0) \quad (35)$$

defined by W_0^{ab} . Hence the fact that (5.3.5) is a perfect duality also means, by the well-known dictionary (cf. [5]) that the kernel of (5.3.7), which is a finite group over k by virtue of 1.5, has a null ℓ -primary component. Saying that this is so for all ℓ therefore means that the isogeny (5.3.7) is an *isomorphism*, i.e. that the divisor correspondence W_0^{ab} is a perfect duality between B_0 and B'_0 . The preceding considerations only show that the kernel of (5.3.7) is a p -group. Let us therefore state the more complete result :

Theorem 0.19 (5.4). *Let S be a trait, and A_K an abelian scheme over S having semi-stable reduction over S . Denote by B_0, B'_0 the abelian parts of the Néron models of A_K and of the dual abelian scheme A'_K respectively. Then the canonical divisor correspondence W_0^{ab} on (B_0, B'_0) , deduced by VIII 4.8 from the canonical biextension W_0 of $(A_0^0, A_0'^0)$ (special fibre of the biextension W of 1.4), is a perfect duality between B_0 and B'_0 , i.e. defines an isomorphism of each of these abelian schemes with the dual of the other.*

The proof of 5.4 in the general case will be given in No. 7, as a consequence of a very natural and important construction due to RAYNAUD. Let us only note here that to prove 5.4, one may assume S complete, in view of 3.3.

5.5. To express conveniently the extreme factors in the filtrations (5.1.10) and (5.1.11), in the case of semi-stable reduction, let us introduce the twisted constant group schemes over S

$$\underline{M}' = D(T_0) , \quad \underline{M} = D(T'_0) , \quad (36)$$

duals of the maximal tori T_0, T'_0 of the special fibres of A, A' (SGA 3 X 5.7). As S is local complete, these group schemes may be considered as the special fibres of twisted constant group schemes

$$\underline{M}' , \quad \underline{M} \quad \text{on } S ; \quad (37)$$

thus, if the torus T_0 is split (for example if k is separably closed) i.e. if $\underline{M}'_0$ is the constant group defined by the ordinary group

$$M' = \text{Hom}_{k\text{-gr}}(T_0, \mathbb{G}_{mk}) , \quad (38)$$

then $\underline{M}'$ is the constant group defined by M' :

$$\underline{M}' = \underline{M}'_S .$$

One may also note that, as T_0 and T'_0 are isogenous (2.2.6), T'_0 is split if T_0 is, so that there is reason to also introduce in this case

$$M = \text{Hom}_{k\text{-gr}}(T'_0, \mathbb{G}_{mk}) , \quad (39)$$

and one will then have

$$\underline{M} = \underline{M}_S .$$

This being posited, one will evidently have, for every integer n , isomorphisms

$$\underline{M}'_n \simeq D(T_n) , \quad \underline{M}_n \simeq D(T'_n) , \quad (40)$$

where T'_n is defined in terms of A' as T_n in terms of A (5.1.2), whence one deduces canonical isomorphisms (where one sets $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{mS})$) :

$$T_\ell(T_n) \simeq \check{\underline{M}}'_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) , \quad T_\ell(T'_n) \simeq \check{\underline{M}}_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) , \quad (41)$$

and by passage to the limit on n canonical isomorphisms :

$$\begin{cases} T_\ell(A^0)^t = T_\ell(\hat{T}) \simeq \check{\underline{M}}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) \\ T_\ell(A'^0)^t = T_\ell(\hat{T}') \simeq \check{\underline{M}} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1) . \end{cases} \quad (42)$$

In the semi-stable reduction case, one concludes by duality, using the orthogonality theorem 5.2 and relation (5.2.2), canonical isomorphisms :

$$\begin{cases} T_\ell(A_K)/T_\ell(A_K)^f \simeq \underline{M} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \\ T_\ell(A'_K)/T_\ell(A'_K)^f \simeq \underline{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell . \end{cases} \quad (43)$$

One finds likewise, thanks to the same formula (5.2.2), canonical isomorphisms :

$$\begin{cases} T_\ell(A_K)/T_\ell(A_K)^t \simeq D(T_\ell(A'_K)^f) \\ T_\ell(A'_K)/T_\ell(A'_K)^t \simeq D(T_\ell(A_K)^f) , \end{cases} \quad (44)$$

where one will recall that the groups $T_\ell(A_K)^f$ and $T_\ell(A'_K)^f$ appearing in the second member are the restrictions to η of Barsotti–Tate pro- ℓ -groups defined over S , $T_\ell(A^0)^f$ and $T_\ell(A'^0)^f$ respectively;

the letter D denotes here the duals in the sense of Barsotti–Tate pro-groups, namely $\mathrm{Hom}(-, \mathbb{Z}_\ell(1))$.

Proposition 0.20 (5.6). *Suppose S complete and that A_K has semi-stable reduction over S . Then for every prime number ℓ (which may be equal to the residual characteristic of S), the pro- ℓ -Barsotti–Tate group $T_\ell(A_K)^t$ is the generic fibre of a Barsotti–Tate group over S , of “toric type”, given by (5.5.7), while $T_\ell(A_K)/T_\ell(A_K)^f$ is the generic fibre of a Barsotti–Tate group over S of pro-étale type, given by (5.5.8). Finally the groups $T_\ell(A_K)^f$ and $T_\ell(A_K)/T_\ell(A_K)^t$ are the generic fibres of Barsotti–Tate groups defined over S , equal respectively to $T_\ell(A^0)^f$ and to $D(T_\ell(A^0)^f) = \mathrm{Hom}(T_\ell(A^0)^f, \mathbb{Z}_\ell(1))$.*

5.6.1. One will note that it amounts to the same for the étale covering S' to split $T_\ell(A_K)^t$ or to split $T_\ell(A_K)/T_\ell(A_K)^f$, for this means respectively that S' splits the twisted constant groups $\underline{M}$ and $\underline{M}'$, which are isogenous, since T_0 and T'_0 are so. One sees at the same time that there exists a smallest étale extension S' of S which does the job.

Let us also mention the following easy consequence of the definitions of 2.2.3 :

Proposition 0.21 (5.7). *Suppose A^0 is ℓ -divisible, S henselian. Let F_K be a constructible ℓ -adic sheaf on K which is “unramified” i.e. which extends to a constant twisted ℓ -adic sheaf F on S , and let $u: F_K \rightarrow T_\ell(A_K)$ be a homomorphism of pro-groups, then u factorises into a homomorphism from F into $T_\ell(A_K)^f$.*

Proof. Indeed, one will have

$$u = (u(\nu))_{\nu \geq 0}, \quad \text{with } u(\nu): F_K(\nu) \longrightarrow T_\ell(A_K)(\nu) = {}_{\ell^\nu}A_K. \quad (45)$$

By the universal property of A , as $F(\nu)$ is étale hence smooth over S , the morphism $u(\nu)$ comes from a homomorphism

$$v(\nu): F(\nu) \longrightarrow {}_{\ell^\nu}A,$$

and by definition of the fixed part, this homomorphism factorises as

$$v(\nu): F(\nu) \longrightarrow ({}_{\ell^\nu}A)^f.$$

As the image of the induced morphism on the special fibres falls into the infinitely ℓ -divisible part of A_0 , one concludes that $v(\nu)$ even factorises through $({}_{\ell^\nu}A^0)^f$. By passage to the limit on ν , one finds a homomorphism

$$v: F \longrightarrow T_\ell(A^0)^f,$$

and the morphism induced on the generic fibres evidently gives the sought factorisation. $\square$

5.8. Using the orthogonality theorem, we shall now obtain the main variants of the Néron–Ogg–Shafarevich criterion :

Theorem 0.22 (5.8). *Let A_K be an abelian scheme over K . Suppose that S is strictly local, and let ℓ be a prime number $\neq p$. The following conditions are equivalent :*

- (i) A_K has good reduction over S , i.e. A_K extends to an abelian scheme over S .
- (ii) The specialisation homomorphism $I \rightarrow \text{Aut}(T_\ell(A_K))$ is trivial.

Theorem 0.23 (5.9). *With the notation of 5.8, suppose A_K has semi-stable reduction over S . Then the following equivalent conditions :*

- (i) A_K has good reduction over S .
- (ii) $T_\ell(A_K)^t = 0$.
- (iii) $T_\ell(A_K)^t = 0$ for one prime number $\ell \neq p$.

Theorem 0.24 (5.10). *With the notation of 5.8 (but without necessarily supposing A_K to have semi-stable reduction), the following equivalent conditions :*

- (i) A_K has good reduction over S .
- (ii) $T_\ell(A_K)^f = 0$ for one prime number $\ell \neq p$.
- (iii) $T_\ell(A_K)^f = 0$ for every prime number $\ell \neq p$.

Proof. (i) $\Rightarrow$ (ii) is clear, for if A_K has good reduction, then $A^0 = A$ and $T_\ell(A^0)^f = T_\ell(A)$, hence $T_\ell(A_K)^f = 0$.

(ii) $\Rightarrow$ (iii) is trivial.

(iii) $\Rightarrow$ (i) : If $T_\ell(A_K)^f = 0$, then A_0^0 is unipotent (2.2.5). But for it to be zero, it is necessary and sufficient that the euclidean dual of $T_\ell(A_K)/T_\ell(A_K)^f$ be zero, which still means that the maximal torus of A_0^0 is zero, i.e. A'_K has good reduction. As $A''_K = A_K$, this means that A_K has good reduction. $\square$

5.11. One also finds criteria for good reduction as well as for semi-stable reduction in terms of finite groups :

Theorem 0.25 (5.12). *Suppose S strictly local, and let n be an integer ≥ 1 prime to p . The following conditions are equivalent :*

- (i) A_K has good reduction over S .
- (ii) The group ${}_n A_K$ is unramified over S , i.e. I acts trivially on ${}_n A(\overline{K})$.

Proof. If A_K has good reduction, ${}_nA$ is finite and smooth over S , hence ${}_nA_K$ is unramified. Conversely, suppose that ${}_nA_K$ is unramified. Then ${}_nA$ is a finite and flat scheme over S (SGA 3 X 5.9), and as ${}_nA_K$ is unramified, the special fibre ${}_nA_0$ is étale over k . Hence ${}_nA$ is étale over S . For $p > 1$, this implies that A_0 is an abelian scheme, hence A_K has good reduction. $\square$

Theorem 0.26 (5.13). *With the notation of 5.12 (but without supposing A_K to have semi-stable reduction), the following equivalent conditions :*

- (i) A_K has semi-stable reduction over S .
- (ii) For every element $g \in I$, the multiplicity of 1 as a root of the characteristic polynomial of the automorphism of $V_\ell(A_K)$ defined by g is equal to 2α , where α denotes the rank of the abelian scheme $B_0^{A_K}$ appearing in the decomposition (2.2.2) of the special fibre of the Néron model of A_K .

Proof. (i) $\Rightarrow$ (ii) : If A_K has semi-stable reduction, then the characteristic polynomial of $g \in I$ is of the form $(t - 1)^{2\alpha}P(t)$, where $P(1) \neq 0$, whence (ii).

(ii) $\Rightarrow$ (i) : This is a criterion of M. RAYNAUD, whose proof will be given elsewhere. $\square$

Theorem 0.27 (5.14). *Suppose S strictly local. Let ℓ be a prime number $\neq p$, A_K an abelian scheme over K , and suppose that A_K has semi-stable reduction. Consider the canonical injection*

$$T_\ell(A_K)^f \hookrightarrow T_\ell(A_K), \quad (46)$$

and let μ be the rank of the toric factor of the special fibre of the Néron model of A_K . Then the following equivalent conditions :

- (i) $\mu = 0$, i.e. A_K has good reduction over S .
- (ii) I acts trivially on $T_\ell(A_K)/T_\ell(A_K)^f$, and the characteristic polynomial of the automorphism defined by an element $g \in I$ on $T_\ell(A_K)/T_\ell(A_K)^f$ has rational integer coefficients independent of ℓ .

Theorem 0.28 (5.15). *With the notation of 5.14, suppose k finite. Then the following conditions are equivalent :*

- (i) A_K has semi-stable reduction over S .
- (ii) $T_\ell(A_K)/T_\ell(A_K)^f$ and $T_\ell(A_K)^t$ are defined over K (i.e. I acts trivially on these groups).

§ 6. Remarks on the construction of the toric and fixed parts in the non-semi-stable case.

6.1. Let us now return to the general situation of § 2, that is, we do not suppose A_K to have semi-stable reduction. One then has, for every prime number $\ell \neq p$, the filtrations (1.3.2) and (2.3.0) :

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K). \quad (47)$$

The fundamental result of 3.6 teaches us that the toric part $T_\ell(A_K)^t$ is zero for ℓ sufficiently large, i.e. if K' is a finite extension of K such that $A_{K'}$ has semi-stable reduction over S' , then $I' = \text{Gal}(\overline{K}/K')$ acts trivially on $T_\ell(A_K)^t$, so that $T_\ell(A_K)^t$ is the generic fibre of a Barsotti–Tate pro-group of *toric type* over S . Likewise, the fixed part $T_\ell(A_K)^f$ is defined for all $\ell \neq p$, and we have seen in 2.2 that $T_\ell(A_K)^f$ is the generic fibre of a Barsotti–Tate pro-group over S , restriction to η of $T_\ell(A^0)^f$, when S is strictly local and A^0 is ℓ -divisible.

We now propose to show how the results of § § 4, 5 allow in their turn to simplify and make precise the structure of the filtration (6.1.1) in the general case, and to construct the toric and fixed parts in an intrinsic way, without passing through the semi-stable reduction.

6.2. Let S be strictly local. Recall (1.4) that we have constructed a *biextension* W of (A^0, A^0) by $\mathbb{G}_{mS}$, where A' is the Néron model of the dual A'_K of A_K . This biextension reduces on the special fibre to a biextension W_0 of (A_0^0, A_0^0) by $\mathbb{G}_{mk}$, and this biextension defines a divisor correspondence, identifying B'_0 with the dual of B_0 (and reciprocally), where B_0 and B'_0 denote the “abelian parts” of A_0^0 and A_0^0 respectively (2.2.2).

Let T_0 be the maximal torus of A_0^0 , and T'_0 the maximal torus of A_0^0 . According to 2.2.6, the twisted constant group schemes $M = D(T'_0)$ and $M' = D(T_0)$ are isogenous. The fact that W_0 vanishes on $T_0 \times T'_0$ (VIII 4.10) means that the pairing

$$T_\ell(T_0) \times T_\ell(T'_0) \longrightarrow T_\ell(\mathbb{G}_{mk}) \quad (48)$$

is trivial, which is evident since $T_\ell(T_0)$ and $T_\ell(T'_0)$ are Barsotti–Tate pro-groups of toric type, and every homomorphism from such a group to $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{mS})$ is trivial. The vanishing of (6.2.1) allows one to pass to the quotient and obtain a pairing :

$$T_\ell(B_0) \times T_\ell(B'_0) \longrightarrow T_\ell(\mathbb{G}_{mk}), \quad (49)$$

which is a *perfect duality* according to 5.4.

6.3. Let ℓ be a prime number $\neq p$. We shall now show that the factors of the filtration (6.1.1) can be reconstructed in terms of the filtration (1.3.2) and of the biextension pairing. Indeed, consider

the exact sequence of Galois cohomology

$$0 \longrightarrow T_\ell(A_K)^t \longrightarrow T_\ell(A_K)^f \longrightarrow T_\ell(A_K)^{\text{ab}} \longrightarrow 0 , \quad (50)$$

where $T_\ell(A_K)^{\text{ab}}$ denotes the quotient $T_\ell(A_K)^f / T_\ell(A_K)^t$, and the dual exact sequence

$$0 \longrightarrow T_\ell(A'_K)^t \longrightarrow T_\ell(A'_K)^f \longrightarrow T_\ell(A'_K)^{\text{ab}} \longrightarrow 0 . \quad (51)$$

The biextension pairing defines a perfect duality

$$T_\ell(A_K)^{\text{ab}} \times T_\ell(A'_K)^{\text{ab}} \longrightarrow \mathbb{Z}_\ell(1) , \quad (52)$$

so that $T_\ell(A_K)^{\text{ab}}$ and $T_\ell(A'_K)^{\text{ab}}$ determine one another by duality.

6.4. We have seen in 2.2.6 that the ranks of the tori T_0 and T'_0 are equal. More precisely, the constant twisted groups $M = D(T'_0)$ and $M' = D(T_0)$ are isogenous. In fact, the divisor correspondence W_0^{ab} allows us to define a *canonical isomorphism* between the $\mathbb{Q}_\ell$ -vector spaces

$$V_\ell(A_K)^t \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \simeq V_\ell(A'_K)^t \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell , \quad (53)$$

and the structure of the factors of the filtration (6.1.1) is completely determined by the biextension pairing.

6.5. To conclude, let us point out that the construction of the toric part in the case where $\ell = p$ (hence in residual characteristic p) is done by the same method as in 5.1 : one defines the toric part by

$$T_p(A^0)^t = T_p(\hat{T}) \subset T_p(A^0)^f , \quad (54)$$

where $\hat{T}$ is the formal torus defined in (5.1.3). This construction is valid under the hypothesis that A^0 is p -divisible, which is the case when A_K has semi-stable reduction over S (3.5). In the general case, one can always define $T_p(A^0)^t$ as the largest sub-pro-Barsotti–Tate group of toric type contained in $T_p(A^0)^f$.

6.6. Let us summarise the main properties of the monodromy filtration in the general case : to every abelian scheme A_K over K , one has associated a canonical filtration :

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K) , \quad (\ell \neq p) , \quad (55)$$

whose successive factors are :

- $T_\ell(A_K)^t$: toric part, of rank $\mu = \dim T_0$, defined over a finite extension of K (semi-stable reduction) ;

- $T_\ell(A_K)^{\text{ab}} = T_\ell(A_K)^f / T_\ell(A_K)^t$: abelian part, of rank $2\alpha = 2 \dim B_0$, in perfect duality with the abelian part of the dual ;
- $T_\ell(A_K) / T_\ell(A_K)^f$: étale part, of rank μ , isomorphic to $M' \otimes \mathbb{Z}_\ell$ for the semi-stable reduction.

The ranks of these groups are related to the invariants of the decomposition (2.2.2) by :

$$\dim A_K = \alpha + \lambda + \mu , \quad (56)$$

where $\alpha = \dim B_0$, $\lambda = \dim U_0$, $\mu = \dim T_0$.

In the semi-stable reduction case ($\lambda = 0$), one has $\dim A_K = \alpha + \mu$, and the monodromy filtration reduces to :

$$0 \subset T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K) , \quad (57)$$

with :

- $\text{rank } T_\ell(A_K)^t = \mu$,
- $\text{rank } T_\ell(A_K)^f = 2\alpha + \mu$,
- $\text{rank } T_\ell(A_K) / T_\ell(A_K)^f = \mu$.